TERRY WINOGRAD

An Oral History

conducted by Richard Fikes

STANFORD HISTORICAL SOCIETY ORAL HISTORY PROGRAM

Stanford University

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Terry Winograd

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Introduction

This oral history was conducted by the Stanford Historical Society Oral History Program in collaboration with the Stanford University Archives. The program is under the direction of the Oral History Committee of the Stanford Historical Society.

The Stanford Historical Society Oral History Program furthers the Society's mission "to foster and support the documentation, study, publication, dissemination, and preservation of the history of the Leland Stanford Junior University." The program explores the institutional history of the University, with an emphasis on the transformative post-WWII period, through interviews with leading faculty, staff, alumni, trustees, and others. The interview recordings and transcripts provide valuable additions to the existing collection of written and photographic materials in the Stanford University Archives.

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Abstract and Topics

Terry Winograd, Professor of Computer Science, Emeritus at Stanford University, describes his early life in Colorado, his education in linguistics and applied mathematics, and his research career in artificial intelligence (AI) and human-computer interaction. Topics include the development of the SHRDLU system; memories of Xerox PARC and the DC Power Lab; his collaboration with Fernando Flores on *Understanding Computers and Cognition*; and the founding of Computer Professionals for Social Responsibility. He also reflects on the relationship between AI, human-computer interaction, and computer science; describes project-based and ethics courses he developed and the students he advised, including the two founders of Google; and discusses his continued engagement in social issues, philanthropy, and family life.

Part 1 **p. 13**

[00:00:00 – 00:30:57] Growing up in Greeley, Colorado • Building a computer in high school • Undergraduate years at Colorado College • Programming a Control Data 160 • Creating programs to calculate isodose curves • Fulbright studying linguistics at the Communication Research Center at University College in London with Michael Halliday • History of using computers to decode languages • PhD studies at MIT with Marvin Minsky and Seymour Papert • Developing SHRDLU, an early use of AI • Importance of context in language • Working with Daniel Bobrow on KRL (Knowledge Representation Language) • Summer 1972 at Xerox PARC with Bobrow • Coming to Stanford in 1973 as visiting professor in Computer Science and Linguistics

[00:31:17 – 00:58:34] History of computer science at Stanford and relationship to the emerging field of AI • Joining Stanford faculty • Memories of DC Power Lab and PARC • Graduate students • Book *Language as a Cognitive Process* • Process of producing camera-ready pages of his book • Reflections on why he moved away from study of artificial intelligence; impact of Stanford/UC Berkeley Story Group

[00:59:10 – 01:33:07] Story of how Fernando Flores, a former professor and a member of Salvador Allende's government, ended up at Stanford • Collaborating with Flores on

Understanding Computers and Cognition; incorporating ideas of Humberto Maturana and Martin Heidegger • Shift toward studying human computer interaction • Description of landscape of Computer Science at Stanford and where AI fits • Teaching undergraduate programming course using ALGOL W and punch cards • History of HCI, including initial focus on constructing computers and cognitive psychology; roles of Stuart Card and Alan Kay at Xerox PARC • Attitudes of other computer science areas toward HCI

Part 2 **p. 63**

[00:00:00 – 00:29:45] Impact of early life in Colorado, including being one of only Jewish families in his town • Early opportunities to explore his intellectual curiosity, including building a computer for a science fair • Thoughts on relationship between computer science and HCI • HCI focus on how computers are used rather than mechanics; importance of engaging with people • Developing project-based courses to teach HCI; projects in Kenya funded by the Nokia Research Center

[00:30:33 – 01:06:03] d.school approach of interdisciplinary teams with T-shaped people participants • Projects in Kenya on identifying water suppliers and joining walking groups for safe transit • Developing course with Helen Nissenbaum on ethics in computing • His early political and social activism • Founding of Computer Professionals for Social Responsibility; early initiatives, including opposition to SDI and concerns about privacy issues with creation of large data bases with personal information

[01:06:04 – 01:46:24] Serving on boards of Corporate Accountability and Bend the Arc • Approach to advising students • Memories of graduate students Larry Paige and Sergey Brin • Sabbaticals at Interval Research and Google • Teaching a freshman seminar on social issues in computing • Decision to retire; activities during retirement; grandchildren

Terry Winograd

Biography

Professor Winograd's focus is on human-computer interaction design and the design of technologies for development. He directed the teaching programs and HCI research in the Stanford Human-Computer Interaction Group, which recently celebrated its 20th anniversary. He was also a founding faculty member of the Hasso Plattner Institute of Design at Stanford (the "d.school") and on the faculty of the Center on Democracy, Development, and the Rule of Law (CDDRL).

Winograd was a founding member and past president of Computer Professionals for Social Responsibility. He is on a number of journal editorial boards, including Human Computer Interaction, ACM Transactions on Computer Human Interaction, and Informatica. He has advised a number of companies started by his students, including Google. In 2011 he received the ACM SIGCHI Lifetime Research Award.

S T A N F O R D U N I V E R S I T Y

PROJECT:	STANFORD FACULTY ORAL HISTORY PROJECT
INTERVIEWEE:	TERRY WINOGRAD
INTERVIEWER:	RICHARD FIKES
DATE OF INTERVIEW:	FEBRUARY 14, 2023
PART:	1 OF 2

Fikes: My name is Richard Fikes. I'm a professor emeritus in Stanford's Computer Science Department and former director of Stanford's Knowledge Systems Laboratory. It is February 2023. And it is my pleasure today to interview Stanford Computer Science professor emeritus Terry Winograd for the Stanford Faculty Oral History Project. Professor Winograd's many accomplishments while at Stanford include founding and directing Stanford's Human-Computer Interaction group and being a founding faculty of the Hasso Plattner Institute for Design. He is speaking to us today from his home on the Stanford campus. So Terry, we've been fellow computer scientists for over fifty years. And I'm delighted to have this opportunity to talk with you about your life and career.

Winograd: Thank you.

Fikes: We are going to spend our time together exploring your background and professional history with a particular focus on your time at Stanford. I look forward to hearing about your personal experiences and decisions that shaped your career, the origins of your career, and all that happened while

you were at Stanford. So let's begin by going back to the beginning. Perhaps you can tell us a bit about where you were born and what your early life was like.

Winograd: It's actually an accident that my official birthplace has almost nothing to do with my life. My father was in the Navy in World War II and was stationed in Washington DC and got married there. And so I was born in Maryland. So that's officially my state of birth. But when I was six weeks old, he moved back to Colorado, where he had grown up, and my mother, who had grown up in Washington DC, came with him. So for all practical purposes, I'm from Colorado even though my driver's license doesn't say that.

[00:02:04] I grew up in a relatively small town—Greeley, Colorado—which was a mainly agricultural community, about 20,000 people when I was growing up. But it also had the Teachers College. So it had a certain academic cast to it for an agricultural town. My father was a businessman. He had a steel business and junkyard. I spent my summers working in the junkyard. I guess I can tell you how to sort different kinds of brass.

I went to the local public high school there. At the time, it was the only one in town. Now, there's more. I skipped a couple of grades. So I actually went to college at the age of sixteen and went to Colorado College, which is a small liberal arts college in Colorado Springs. I'm really glad that I did that instead of going to a big university or technical school. At the time, there was good financial aid—I got a scholarship and all that—but it was an opportunity to really branch out. I was officially a math major. I took one math course every quarter, every semester, I guess.

I was editor of the college paper. I took courses in poetry and political science and linguistics. So it gave me a chance to really see that there was a lot more than just the technical stuff, even though that was my strong point. When I was a little kid, pre-kindergarten, somebody said, "What do you want to be when you grow up? What do you want to do?"

And I said, "I want to be in the arithmetic business because I love arithmetic." So from the beginning, I was headed that direction.

Fikes: What was your first contact with computers?

Winograd: [00:03:56] My first contact with computers of a sort was my high school science project. I built a little computer that did a particular calculation. It wasn't transistors and that kind of stuff. It was switches and resistors and so on. So that was when I was about fifteen, fourteen whatever it was, before I graduated that I actually built a "computer." As I say, it's not the same kind. Then when I went to college, there was a very brief encounter during a period which we had at Colorado College. We had two semesters, and in between them there's a week of independent study programs they call the Symposium. The year I was, I guess, a sophomore, or a junior, they brought in a small computer, which nobody had ever seen before, of course.

It was sitting there in the library atrium. And you could try it, playing, do things with it. I learned how to program a little. I wrote a little program that played mancala, if you know the game where you move rocks around a circle. So that was sort of a little experience. It was a Control Data 160, which looks like a big office desk. So it was a sort of a personal computer, even though in those days, it was not personal. And the input-output was all

paper tape. One of my great adventures was—it was hot that day and they had the fan on. I ran the paper tape into the paper tape reader. You can guess, it went right straight through into the fan. [laughter]

[00:05:41] So that was I/O experience. Then later in that year, a doctor who had become head of radiology at the local hospital in Colorado Springs who had previously been in a big medical center in Denver. He had a research project—he was a radiologist—to calculate radiation doses for cancer therapy. He had had a programmer working with him when he was in Denver. He arrived in Colorado Springs, and he called up the college. He said, "Can you send me a computer science student?"

And they said, "A what? We don't have computer science. How about a smart math student?" So I ended up getting a job at the hospital with Dr. Perez Tamayo, was his name, and wrote programs to calculate isodose curves, they're called. If you plot the radiation on a certain path that it takes, how much radiation will be at each point? And of course, the basis of those is to get a lot on the tumor and not much anywhere else. I didn't know anything about computers: I programmed by pushing the buttons on the console. I programmed it in raw machine language. I didn't know anything about compilers. I sort of learned from the bottom up, but it was mine. I was the only person using it. I could go any time, night or day, and play with the computer. It was my personal computer.

Fikes: So the work that you did for the hospital research project was done on the 160 at the college?

Winograd: At the hospital.

Fikes: At the hospital?

Winograd: It was the same—it turned out to be the same model of computer.

Fikes: I see. After getting your BA at Colorado College, you went to University College in London for a year on a Fulbright scholarship. Tell us about your choosing to go to London and studying linguistics there.

Winograd: [00:07:30] So it's kind of an accident, a fortunate one. I had always been interested in language. It's just something I was curious about. There was no linguistics at a small college like Colorado College. I went to the anthropology teacher, and I said, "I'm interested."

And he said, "Why don't we do an independent reading course on linguistics?" So I did an independent reading course on traditional linguistics, just out of curiosity. It wasn't my main direction. My main direction was technology and math and computing. It was just a side thing I was interested in.

The campus Fulbright advisor said, "Why don't you apply for a Fulbright? You're a very good student. And they have geographic quotas, which means they're going to send a certain number from Colorado. So you're not competing with those Harvard people. You're competing with other colleges in Colorado."

So I said, "Well, that sounds fun. But I have two problems. Problem one is I'm interested in computing, and nobody outside the US really does computing." That was chauvinism because in fact, computing started in Britain with Turing. But I didn't know that. And the second problem, I said,

"The only foreign language I've taken in college is Russian, and you can't get a Fulbright to Russia." We were stuck in major Cold War at that time.

And he said, "Why don't you look around?"

So I said, "Well, I'm interested in language." And it turned out there was something officially called the Communication Research Centre at University College led by Professor Halliday [Michael Halliday]. They weren't doing computers. But they were interested in language from a more computational point of view. So I said, "Well, it would be great to go to Europe for a year. So why don't I go there, study linguistics, then I'll come back and get serious in computing?"

[00:09:20] I had actually applied and been accepted to MIT. I was headed to go there when I got back. So I went off for a year to London, which was great. I did go to Europe. I took trips to various places, including Eastern Europe and Russia. I had a course in Swahili, which I remember almost nothing about, but I did take the course in Swahili. That got me more connected to thinking about language and linguistics. So when I came back to my position at MIT, that was sort of the direction I was interested in going. It was just when people were starting to use computers for things other than traditional calculations. I said, "How about using it for language? That sounds like something fun to do."

Fikes: I wanted to ask you about that because at that time, there was no computational linguistics as far as I know. You said in your application for the Fulbright, "Language is important. And computers can do it. And maybe we should try to connect them." So in today's world, that connection

between natural language and computers is commonplace and pervasive. To what extent was that a new thought in the 1960s?

Winograd: Well, no, it had been around for a long time. There was a guy, Bar-Hillel [Yehoshua Bar-Hillel] was his name, who back in the early 1950s talked about computer translation. He said, "Look, translation is just coding."

Think of French as a funny code. We could use computers because, of course, in the war, Turing was doing decoding. That was the whole big deal with creating computers. Why can't we decode French as well as we can decode the enigma machine or whatever it is? IBM actually had a program doing machine translation. I remember the World's Fair in what must have been the late 1950s, early 1960s. They had a display with their translation machine.

[00:11:24] In those days, it was very crude because what you could do with the computer pretty easily is word for word. Here is the word in English, here is the word in Russian or whatever it is. And you switch them. The idea of using computers for language and translation, was the thing. It was practical. If you could do it well, you could make it useful in the world. This had been around, as I say, for quite a few years. I think that combining that with what you might call understanding, which is not just to produce a translation but actually use the language in terms of what it means, that came along later with AI and with the idea that you needed some kind of intelligence to do that.

Fikes: Your work was seminal in showing that a computer could understand what was being said in natural language.

Winograd: You're right. Just at about the time I was doing it, ARPA [Advanced Research Projects Agency] was starting up programs on speech. So I did not do that. Everything I did was typewritten language. It wasn't speech. But I think there was also the idea that makes the obvious point. If you're doing something out there in the military field, you don't want to stop and type. You want to say what you want and have the machine understand it. There were speech programs at a number of universities that started just about when I was doing my dissertation.

Fikes: So you mentioned MIT already. So after getting your master's—you've got a master's degree in linguistics.

Winograd: No, I didn't actually.

Fikes: Oh, you didn't?

Winograd: I was in a one-year non-degree program.

Fikes: So you spent a year at University College and then you went to MIT as a PhD student?

Winograd: Correct.

Fikes: So tell us about your choosing to go to MIT. And then what was your experience there as a student?

Winograd: [00:13:25] [sighs] When I reflect, I often say I never really made choices. I just sort of knew what was going to be next. The reason I got into MIT was sort of a fluke because they don't go looking for kids at Colorado College. I had very little technical background. I had one math course a semester, but I scored very, very high in what's called the Putnam Mathematics challenge or test [William Lowell Putnam Mathematical Competition], whatever it

was. They said it's like national honor merit, whatever. I think it's a nationwide test. Anybody, anywhere can take it. It's basic mathematical ability but at a level above high school stuff, so higher level. I got a score which you don't expect from somebody coming from that kind of college.

[00:14:14] So as a result of that, MIT was interested. That's how I was able to get in. I wanted to do computers, and if you said, where are there good universities doing computers? Stanford wasn't at that time yet. It was starting. I didn't want to go to Pittsburgh. I know you went there.

Fikes: Good choice. [laughter]

Winograd: So it's like, well, where else is there? That's where I'll go.

Fikes: Tell us about your experience there as a student in the MIT world with Minsky [Marvin Minsky] and Papert [Seymour Papert]. It was fascinating, I would think.

Winograd: MIT was stuck in like a feudal structure. You had lords of their own domains. They had pretty much independence to do what they wanted. It didn't have a lot of apparatus to say students have to do this, students have to do that.

[00:15:11] They said, "You're a student with the lab there. They'll tell you what to do, go do it." So I was actually not able to get into electrical engineering. Minsky, who was my de facto main advisor, was in electrical engineering, but I had absolutely zero engineering background. So I ended up in applied math. My advisor officially was Seymour Papert, who was in the math department as opposed to electrical engineering—there was no computer science. It was all those departments.

Minsky ran the lab like, "This is my lab. I'll do what I want and do fun things." And of course, you can't imagine a better environment for doing fun things than the students who go to MIT where you've got this world of brilliant hackers who want to try things and want to do things. I think mainly his view was, let these guys—and they were almost all guys—do it, play with these machines, and see what comes out. So that was the atmosphere really.

[00:16:16] The AI Lab was on the ninth floor of a building in Tech Square. It was basically a bunch of mostly undergraduates—graduate students, too, but a lot of undergraduates—trying to see what they could do with these machines.

Fikes: Did you take any courses?

Winograd: Yeah, I took courses. To get my PhD, I had to take some number of math courses. I remember struggling. I said to somebody—and I've forgotten where—I got to MIT as a graduate student and I got in these courses, and the sophomores knew a lot more math than I did.

Fikes: Yeah.

Winograd: Right? [Fikes and Winograd laugh] So I had to really struggle to get through my partial differential equations, for example. I remember that course. And I took a course in algebra, which of course it's algebra in the advanced sense.

Fikes: Abstract algebra.

Winograd: Abstract algebra—they call it abstract nonsense, that was the name for one of the things. Injections and bijections and a whole bunch stuff. So I had to

do that. But that was pretty much a matter of just, okay, get through those courses. It was not a major part of my time. And then in the lab, basically Minsky and Papert said, "Well, what do you want to work on? What do you think is going to be the next interesting thing to do?" And so since I had done the linguistics and I was interested in that, that was an obvious thing for me to work on.

Fikes: It doesn't sound like you got much guidance from Minsky or Papert. Did you work with other students? Or was this pretty much an independent project?

Winograd: It was pretty much an independent project. There were parts of it that were collaborative. First of all, they were developing the underlying computer systems, the MacLisp system and the PDP-6. So in some sense, you're always collaborating with the guys who were not mainly graduate students who are building all those things. For one part of what I did in my dissertation, I needed a program that could do simple kinds of reasoning. Carl Hewitt, who just recently died, was a graduate student working on reasoning systems. He had something called Planner, but he was the kind of guy who if you'd say, "Well, Carl, I'd like to use Planner in my part of my program," he'd say, "Oh, that's great. Well, I'm working on this. I'm working on that. I'll get to it; I'll get it done. I'll get it done." On and on and on. Carl wasn't a finisher in that sense.

[00:19:01] So I got together with two other grad students, Jerry Sussman [Gerald Sussman] and Gene Charniak [Eugene Charniak], both of whom wanted to use it for what they were working on. We did a really

stripped-down version. We called it Micro-Planner because it was so small. We just had the basic nuts and bolts that were needed for our projects. So the three of us did that together.

Fikes: Well, okay. So let's talk about what you produced because it made a great impact. So you completed your PhD at MIT in 1970.

The system you built for your PhD thesis project, SHRDLU, received a great deal of acclaim as a demonstration of artificial intelligence and was the basis for you authoring both a monograph-length publication and a book. Tell us about the origins and development of SHRDLU.

Winograd: So I started off to develop a program that could take in ordinary language sentences—I said written, not spoken, but ordinary language—and respond in a way which required understanding what was in there. So the first question was, what should it be talking about? I sort of strongly believed that in order to make the computer able to reason well, it needed to be a very limited domain. Actually, one of the things Minsky suggested to me, he said, "Why don't you do children's stories because they're simpler? They're children."

[00:20:25] And my view is, no, that's not true. Children's stories depend on this huge body of social and embodied knowledge that kids have by the time they're old enough to read. So they're not simple. They're simple in some sense, like the linguistic part. The sentence structure is simple, but the concepts are not. He did convince Gene Charniak to do that. So Gene Charniak did a dissertation parallel to mine on understanding children's books. I said, "Let me take something that's simpler, really

simpler." And in the lab—because this is the AI lab—there were all sorts of projects, including robotics. One of Minsky's favorite hobbies was robot hands and stuff. He actually designed the robot hand that you see in the movie 2001 [*2001: A Space Odyssey*].

Fikes: [laughs] No kidding.

Winograd: And so there was a project—several projects, but one major one—with a big arm that could go around. It was on a tabletop. And it had these blocks. And they were doing vision to try to see where the blocks were and what they were. Of course they had all the motion planning and stuff to try to go and do things.

[00:21:34] So I said, "Why don't I take that as my domain?" So you can tell that to go pick up a big red block, which is the first line of my demonstration. You type that in English, and it figures it out and it does it. However, just like Planner, that block-moving stuff was always in process. They were hard problems. Vision was a hard problem. The motion planning was a hard problem. I did not want to put my progress behind somebody getting the arm to work right. So I said, "I'll make up a fake block world."

So that's what I did. I made up on a computer display, which in those days you had to program at a very low level, a computer display of what that robot arm was doing simplified. And that was what the program talked about. It dealt with the world of pseudo-blocks in this computer display, simulated world. Metaverse, right? It wasn't real blocks.

Fikes: Right. Well, it seems like those were a very important set of choices that you made and that reflects a lot of the problems actually that artificial

intelligence has had over the years. You could drastically simplify the world, as you said, to produce interesting results. But it's very easy to think, well, given that we can do that, then, you've got to be able to do children's stories or any kind of natural language understanding. I don't know if you want to make any comments about that.

Winograd: Let me just make one other comment on why it became so notable as well. Because I had studied linguistics, I had thought about things like linguistic context. So what you say in a given sentence is not just what's in that sentence. It's where it fits into a longer dialogue. And to pick up the most obvious example, pronouns. If I say, "Put it down," what is it? You have to know what "it" is because in some previous sentence, you said, "Pick up the big red block." If I say, "Get another one," and so on, and so on.

[00:23:51] In our normal language, we just assume that everything we say is going to be heard in the context of what has gone before and what the other person knows. I put in, again, simplified versions of that but enough that it gave it a natural feel. I think that really got people's attention. There was not that sort of mechanical step-by-step programming sense of it, but that you are really talking to somebody who understood a dialogue. I think that was one of the pieces of it that made it notable.

The other part you asked about is the simplifications. And I was certainly, at that time, one of those people who said, okay, we'll start this. We'll get more complicated, more complicated, more complicated. Although I had an appreciation for how language was not as simple as logical deduction, I figured we'll figure out a way to get around that. And in

particular, can the reasoning programs deal with the kind of reasoning that people do, which involves analogy, involves all sorts of things that aren't simple A implies B logic? That's how I got engaged working with Danny Bobrow [Daniel G. Bobrow] on something we called KRL for Knowledge Representation Language, which you may get into later.

[00:25:13] But the idea was, can you create reasoning languages which are more like natural human reasoning and less like simple logical deduction? I was at the time pretty optimistic—if you'd asked me. I was not a flag waver. Minsky loved to say things like, "Well, another ten years and it will be really smart," and so on, and so on.

But I was, "Okay, I don't know. I don't know anything. It's hard. But I believe we can work on this step by step and get there." It took me a while before I stopped and reflected on that.

Fikes: You mentioned Dan Bobrow, just to keep the chronology going here. After completing your PhD, you stayed at MIT. I guess you were an instructor and then an assistant professor, right? My understanding is that you spent a lot of that time continuing to work on SHRDLU and you wrote the monograph and the book.

[00:26:16] You can talk about that. But since you mentioned Bobrow, in the summer of 1972 you came out to Xerox Palo Alto Research Center, known as PARC, which is where Bobrow was, and then you were a consultant at PARC multiple times during the next eleven years. So tell us about your choosing to first come to PARC.

Winograd: So let me take a step back on Danny. He was my predecessor at the lab. He was Minsky's student at the AI Lab. He finished a couple of years before I got there. I've forgotten the exact dates. He had a program called Student, which you typed in algebra problems, word problems. So, "Mary is twice as old as Sally was when Sally was four years younger"—those kinds of problems. Or, "Danny has six apples more than da-da-da."

[00:27:04] You type those in English, and it converted them to algebraic equations and solved them really. This is algebra in the grade school sense, or the middle school sense, very simple problems. But again, it was one of the very first programs that you interacted with in English, in ordinary language. That was the background when I went there. That was the program that had been done before. It was not a dialogue. There were various ways in which it was more simplified.

Of course, it knew nothing about apples or ages. It just converted them all to equations. And so I knew him. He would hang around the lab a little bit. I went and visited him at BBN, Bolt Beranek and Newman, which is where he was working then. He and Warren Teitelman were building a LISP system, and I was using LISP for my dissertation at MIT. It was a different LISP. It was BBN LISP instead of MIT-LISP. But there were a lot of things in common we could talk about. And then he got hired from BBN to come to PARC and to set up a natural language group at PARC. And so he said to me—and I don't remember the exact timing or sequence. He said, "Why don't you come out and do an internship there? And we'll see what's going on." So that was the summer of 1972?

Fikes: 1972, I think.

Winograd: I came out for a summer at Xerox PARC to work with him. And then when I ended up coming to Stanford, of course, it was easy to make a much stronger connection and become part of that group.

Fikes: Right. Well, you came to Stanford in 1973 as a visiting assistant professor, I think, jointly in Computer Science and Linguistics. Tell us about your choosing to come to Stanford.

Winograd: At about that time, 1972, I was already here for a summer. And then my wife graduated from medical school in Boston, where, of course, she had been when I was at MIT, in 1973. She was very interested in finding a place to go that had a politically oriented family practice program—socially oriented, I should say, not politically. It was family practice but looking at the whole situation people were in, not just at their diseases. That was not particularly common in those days—it's still not common enough—but it was not common at all.

[00:29:42] So we went around the country looking; we didn't travel around the country, but she looked around the country to see where there were programs that fit her idea. One which she thought would be very good was in Miami, Florida, at Jackson Hospital there. So we went to Miami. And I went to Florida International University and talked to them about, "Would they like to have a computer scientist ?"—because they didn't have any. They thought it would be interesting. If we decided to go there, I think I could have probably gotten a job there. Who knows where my career would have gone?

Then we came to San Francisco, to the San Francisco General Hospital program, and she really liked that one. It was great. And of course, we had already been here for a summer the year before. So I came to Stanford and said, "My wife would really like to be in this area for three years. Is there anything we can work out?" If I could have been a visiting something or other for three years, that was fine. Our deal was it was her turn to get her training. I'd been at MIT, which was the best possible for me, so for three years, we'd be wherever she wanted, and then we'd decide. That was sort of the premise, that it was a three-year commitment. And then we'd see.

I came to Stanford. I remember the person I mostly dealt with was Ed Feigenbaum [Edward Feigenbaum] since I was in AI, and he was interested. McCarthy wasn't interested in recruiting type stuff. He was interested in the theory.

[00:31:17] I came, and they said, "Well, it's interesting because we have an assistant professor who is in exactly the same line of work you are. He does natural language understanding in AI. And he is taking a sabbatical next year. So why don't you come as "acting" assistant professor in his slot? And then after that year, we'll see what works out." Roger Schank was the name of the guy. And so I said, "That works for me." So I came here as an acting assistant professor for a year.

He had a joint appointment in Linguistics and Computer Science because he was working on language and computing. And at that point, actually, Computer Science was in the School of Humanities and Sciences.

It wasn't in Engineering. So crossing over with another part of the Humanities and Sciences wasn't as complex. So that's how I arrived at Stanford in Computer Science and Linguistics.

Fikes: My understanding was your office was in John McCarthy's lab. Is that correct?

Winograd: Right. AI was divided into two fairly independent fiefdoms. One was McCarthy's. And that was up at the lab in the hills called the DC Power Lab, which had nothing to do with DC power. It was the name of a person. It was sort of a hacker's fantasy, isolated building up on a hillside. You could do what you wanted. McCarthy was a very nondirective director. He officially was in charge. Really, Les Ernest [Lester Ernest] ran the place. But he said, "Oh, that's interesting. Yeah, sure do that." I mean he had no plan for a bigger thing. He had his own work.

[00:33:20] But they got a lot of money in from ARPA. And he said, "We'll use it for—." They had robotics. They had roving vehicles. They had—I guess this is the ancestor of the Tesla, right? It was Shakey. Well, it wasn't Shakey. It was the Cart. And so it was easy. That was where Schank had his office. So it wasn't a question—I would just sit in his office.

Fikes: I think this is important for the history of Stanford and in particular the Computer Science Department at that time. So having a lab way up in the hills, completely off-campus seems like a very unusual setup. Do you know about the origins of that? How was that for you to be up there, off-campus and, as you say, in this kind of hacker's fantasy?

Winograd: I don't know the history before I got there. I can speculate on it. Let me go back to a bigger picture of computer science. Computer science at Stanford started as a branch of mathematics. So you had people like Dantzig [George Dantzig] and -Forsythe [George Forsythe]. And they were mathematicians, applied mathematicians. But they got into computers for obvious connections. And so the computer science effort sort of grew out of thinking it was a branch of mathematics.

[00:35:00] They brought on people like Knuth [Donald Knuth], who was interested in real computing except from the theoretical algorithms point of view. He wanted to do essentially mathematical theory of what computers were doing. And Bob Floyd, who was doing computer language theory, but, again, not engineering as much as the theory of it. And so there was a sense that that was "real" computer science. There was also electrical engineering, where people were building circuits and stuff. But that was engineering. That wasn't computer science.

Then AI came along. And AI came along with a bunch of money because that was during the period when ARPA was saying, we need to put a lot of money into AI, be the future, and so on. We could talk about the whole history of ARPA funding. They were able to get McCarthy to start his own lab. He was at MIT actually. I think this had already happened before I got to MIT, but I certainly saw him there at various times visiting.

I think there was sort of the sense of, "Let those guys go do their stuff. We don't really think it's computer science, but they're getting money,

and they can build a lab and do a lot of hacking. Why not? Let's let them do it."

[00:36:19] I think there was a kind of hands-off feeling between the department, meaning the head people who had founded it, and AI, which was seen as somewhat flaky in general. And then Ed Feigenbaum came, and he said, "Okay, I can do a different kind of AI. Expert systems." They didn't call them in that day but—

He also was not in the main departmental buildings. He had his own empire somewhere else. So I think there was a kind of distancing feeling. What it meant, I think, primarily is almost no connection to classroom teaching. If you had an office way up there on the hill to bike or drive down from there, park if you drove, go to a class, and then when you were done, if you wanted to go to your office, you went back. It was very detached.

As a faculty member, you have responsibilities for classroom teaching. But I think the sense was your real job was working with the graduate students who had offices up there with you on the hill. And yeah, you went and taught once in a while.

Fikes: You had commented that you spent a lot of time at PARC. I think you wrote, "Stanford was a home. Stanford was students. But it wasn't really an intellectual center for what I was doing."

Winograd: [00:37:41] That's part of the culture of the AI Lab. And it was true at MIT also, as well as Stanford, which is, you have independent researchers who have their projects, and each one of them sort of runs the project with a bunch of graduate students.

Collaborate? Uh-huh, yeah, you share a computer system. And you collaborate on getting that working. But it wasn't like your research—the whole idea of collaborative research was sort of like, well, you can't be the prime author on all the papers if you're not the one person doing it. Academics has this in general. But I think that was part of the culture. So there was nobody else there doing anything on language. Raj Reddy had done some stuff on speech, but I think by then he had left for Carnegie. So it wasn't like there was anybody there who was interested in talking about the things I was working on, any faculty.

[00:38:39] On the other hand, I had come out here to be with people who *were* interested in that at PARC. So I think that was just an easy thing to do. And there was no accounting for why are you not spending your time in your office here, and your office at PARC, and so on. So it was just an easy thing to do. Again, no big decision, right?

I could go down to PARC and had an office there. And I could talk to Danny and to Don Norman and to Ron Kaplan and all sorts of people who were interested in language. So I did.

Fikes: Did you have students at that time?

Winograd: I had students. I had a certain number of inherited students. So as you can imagine, when Roger—oh, I didn't finish the story. But the point is obvious. This one-year sabbatical at Yale turned into a full-time job at Yale for Roger Schank. And he never came back. And my acting, they said, "Well, I guess we'll un-act it. And you'll be real, not just acting. You might

as well stay." So no big drama, it was like, okay, I had that position. He didn't come back. There I was.

[00:39:41] So I had probably three or four students—he had a few students who went with him to Yale. And others, who for various reasons, didn't particularly want to leave Stanford for Yale, I became their advisor. So they were already working on whatever projects they were working on, and I was advising them.

Dick Gabriel, David Levy, Paul Martin. So, I had graduate students, but it wasn't like I had my project at Stanford and the students were enlisted in my project. It was much more they were doing things all related to language, because that's how they had gotten started, but pretty much each one on its own. Just to put it in context, my pattern for the following forty years has been mostly students working on individual projects that were their own as opposed to, "Here's the plan that I need to get done. You take this part. You take this part. You take that part." Just not the style that I had.

[00:40:48] So I did have graduate students. And David Levy, in particular, ended up spending a fair amount of time at PARC, he was engaged with that project. Larry Masinter was another grad student who ended up at PARC. He ended up working on the Lisp systems, not on language, but I was his advisor. So I had some students, but in this kind of fragmented sense of each working on related projects.

Fikes: You have said that in the late 1970s, you spent most of your research time for five years writing a book on natural language processing. It was

published in 1983. That was *Language as a Cognitive Process*. So tell us about the writing of that book and its significance to you and to the field of natural language processing.

Winograd: In hindsight, I can be pretty critical of it. At the time, it was going to be the bible. And again, the model there was Don Knuth who wrote the bible. His *Art of Computer Programming*, which was going to be seven volumes. I think he's still on four now after forty years.

Fikes: Yes, right.

Winograd: When I went to the publisher and I said, "Oh, well I have too much to put in one book. I think I'll do it in three volumes." He said, "Yeah, that's what Knuth said." [laughter]. It was in part of that optimism that we were going to, quote, "solve" natural language, have the basis. And it was all going to fall into place step by step. And then there'd be a bible about how do you do it. And the only part which had really fallen into place at that point for me was the syntactic part of it—not that I knew everything about syntax, but it certainly was pretty manageable. So I said, "Well, why don't I start with a volume on syntax? And then I'll do semantics, and pragmatics, the classical linguistic division."

[00:42:55] So I started writing. And then as often happens—this also happened to Knuth, to compare—you just end up saying, "Well, gee, it's interesting how books are typeset. Maybe I'll do something on typesetting." So, I used his programs. I didn't go into typesetting at the level that Knuth did, but I decided to do the whole book myself. I produced single-handedly basically camera-ready copy for the publisher, and that took a lot of effort.

I did a lot of hacking to do that. We had a primitive, what they call the XGP, Xerox Graphic Printer, at Stanford. It was when they were first doing laser printers at Xerox, and there was one that was available at Stanford. And so I was using that. Then Knuth actually set up a photographic typesetting device in the basement of Margaret Jacks Hall after that. So I spent a lot of time on things which when I look back at it, I say, "Well, that wasn't really the core of where I needed to be."

[00:44:04] I piled together huge thick notebooks of stuff for the later volumes. But it didn't gel. It wasn't like I came through and said, "Okay, now I've got the answer here. Now, I've got the answer there." It was like, "Well, I've got lots of stray stuff, but I don't know what to do with it."

When I said I was mostly engaged with that, I was doing a lot of other stuff—at Xerox, the work on KRL and things like that. But I didn't have a separate research project at Stanford. Sometimes when I was at Stanford, I was working on the book.

Fikes: The typesetting thing seems like a complete diversion. What led you to spend time producing camera-ready copy of the book?

Winograd: It was fun.

Fikes: It was fun. [laughter] Fair enough.

Winograd: I've never been a kind of keep your eye on the purpose kind of person. "Oh, that looks good. That's interesting." A later project, when I was working on what we called the iROOM—this is now into the 2000s—I ended up writing my own little database system in C because I had never learned to program in C. And I thought, that'd be really useful to know. So

again, another rabbit hole down the side. I spent a lot of time learning how to program in C, and I wrote a pretty good program. But it wasn't mainstream. I've always been—if you know the Myers-Briggs typology—a more perceptive than judging, more exploring than finding the goal.

Fikes: You talked about having lots of notes for the future volumes, which never got written. Natural language has proved to be extremely difficult in lots of senses. What was your reaction about the fact that you didn't do the second and third volumes? It wasn't gelling, as you put it. What's your sense of that?

Winograd: There's one view of cognition that language is included in cognition—*Language as a Cognitive Process*, the name of the book—that it's sort of conscious thought just invisible. So when I have to come up with the next thing I say, although I don't, in my brain, say, "Okay, now, this would be this and this would be that," it must be going on somehow unconsciously. And that was the basic view of representational artificial intelligence from the beginning. It was Minsky and McCarthy and so on.

[00:47:09] The real essence of intelligence and cognition was something like articulated thought. And there were disagreements. McCarthy said it was like real logic, formal logic. Minsky said, "No, it's not like logic. It's like algorithms." But in every case, it was something which had a kind of representation and manipulations on those representations and the underlying structure. Then the question is, "Okay, so now, how do you represent things which are not simple logic?"

The example I always used to use in talks was, let's take the word "block." In SHRDLU, you said, pick up a big red block. What did "block" mean? Well, it was very simple. There were three kinds of objects on the table. There were pyramids. There were boxes and—I think there were spheres. Anyway, there were blocks. That's what it meant. It was one of those. Now, if I say to you, "What's a block?" Well, it's more complicated. If I say, "Let's take a walk around the block," obviously it's not the same kind of block. It's meaning is different. Okay, now, there's multiple meanings. So there's block as in thing on the table. And there is block as something like geographic areas to walk around.

[00:48:26] Then, what about if I say, "I don't want the car to roll. So I'll put a block behind the wheel." Now, that's neither of those two kind of blocks. If I said, "Show me a picture of that kind of block," it could be a rock, it could be a plank—it has to do with the fact that it's going to keep it from moving. So you say, "Oh, well, block has this whole field of meaning which has to do with my mind, I can't think of that word, I'm blocked. I have a mental block." What's a mental block? It's nothing like a SHRDLU block.

And so as soon as you start to dig into that, what you realize is that people unconsciously are always using language in ways that don't fit simple definitions and ways of reasoning from those definitions. They're more akin to things you might call analogies, more akin to metaphors. There's a whole world of ways of thinking about language which are not cut and dry like the logical forms.

So now as an AI person, you say, okay, how do we build that? If we admit that it's different from the kind of logic you have with "block", with formal logic, how do you put it into a computer? And that was really the path that Danny Bobrow and I were going down with KRL. Could you have a representation language which was still computable representation so the computer could do it and which had more of this open-ended flexible structure that we see as soon as we look at what people really do with language?

[00:50:02] That was the direction that was going to solve that problem. And I think the real answer to that is that never worked. We built it, and it was useful for some sort of things. If you say, what's the role of KRL in the world of AI? Well, it inspired things like KL-ONE and things like that, but nothing big. And if you look at today's AI, it's a historical footnote. It's nothing current. I think what happened is it just began to sink in that these additional mechanisms and additional gimmicks to logical representations weren't getting close to what happens when people just use ordinary language. So I didn't have an answer to, how do you build that in?

Fikes: So you walked away. Let's talk about your moving away from AI. I want to say some things about this because this was a very big deal. I guess in 1979, you became an associate professor at Stanford. About that time, you made a major decision in your career—namely, you walked away from the field of artificial intelligence. You published statements like, quote, "We argue contrary to widespread current belief that one cannot construct machines that either exhibit or successfully model intelligent behavior." At that time,

you were a prominent member of the AI research community, had been for many years, and you were in one of the top AI labs in the world. So your decision attracted a great deal of attention interestingly enough. For example, in the *New York Times*, the *Times* technology writer, John Markoff, referred to you as, quote, "The first high profile deserter from the world of AI."

Winograd: Was this a contemporaneous quote from Markoff or in his later book?

Fikes: I'm not sure. It was in an article about this transition.

Winograd: Interesting. It may have been when he wrote his book. But in any case, yes.

Fikes: Fair enough, okay.

Winograd: I've known John for many, many years.

Fikes: [00:52:19] My point is simply this was a big deal in the field for you to make that change. I have one more thing to add to this. "Your argument was not calculated to win favor at Stanford where McCarthy's lab was acknowledged world leader in AI. 'Terry was, of course, the bad boy,' recalls Reid Hoffman of LinkedIn, who was one of your students at that time. And Hoffman continued, quote, 'I had conversations with John McCarthy at the time, where John said, you know, Terry's totally crazy, gone off the deep end, needs to be on his meds, et cetera.'" Right? [laughter]

Winograd: It's interesting. Until this interview, I'd never heard those quotes.

Fikes: Well, there you go.

Winograd: I'll have to talk to Reid and get more of that story.

Fikes: Right, right. But talk about that decision. This was a big deal.

Winograd: Yeah, let me go back a little bit before that year, 1979. Part of what was going on during the time I was consulting at PARC and doing stuff there was we had had some connections to the Linguistics Department at Berkeley. George Lakoff and—I don't remember his name, the other guy. Lakoff is a very interesting guy. His basic point was language is metaphor. He wrote a book called *Metaphors We Live By*. So that resonated a little bit with the problems like my example of the block and so on. How do we think about metaphor in terms of language?

[00:53:55] There was a group of graduate students from Stanford and Berkeley who had gotten together to discuss these things. One of whom was David Levy, I mentioned earlier, who was also consulting in our group at PARC. One of whom was Deborah Tannen, who became very well-known later on; she's a Georgetown professor in linguistics. She was at Berkeley. These are people who don't care about the Big Game. These are people that said, "Ah, interesting students at Berkeley, interesting students at Stanford, let's get together." So they started a group they called the Story Group because they were looking at the structure of stories and how that worked. Wally Chafe was the other linguistics professor there. So they were meeting here and there in alternating places. I went to some of those meetings, and then some of the faculty got involved.

I was talking with the faculty there and also with some of the philosophy faculty who got engaged, which is John Searle and Bert Dreyfus. They were pushing a very different direction on AI. And let me just go quickly back in history with Dreyfus. Dreyfus was at MIT when I was there,

but not in the AI Lab. He was in, I guess, the Philosophy Department and wrote a very critical paper—later, it became a book—on AI, *What Computers Can't Do*, and was a kind of a whipping boy for the AI Lab.

[00:55:35] My advisor, Seymour Papert, wrote a paper, called *The Artificial Intelligence of Hubert Dreyfus*. It never got published, but I have a copy of it somewhere in the archives. So I had known of him, but I'd never really spent time talking to him. He was on the other side of the barrier at MIT. But he was interested in AI. And Searle had gotten interested in AI. He was an analytic philosopher student from Oxford. He'd gotten high credentials. He was interested in AI, again, from a critical point of view. So we started having some meetings with the two of them. We would go up to Berkeley, and Dreyfus and Searle will be there, and Danny Bobrow from PARC, and some of these students—Henry Thompson, who's now at Edinburgh, has been there for many years.

It just started a kind of thinking, a kind of confrontation of ideas in a good sense of confrontation. Nobody was hostile. It was like, "Let's really think about how this works." And they had different points, and this often gets confused by a lot of people. But Dreyfus's point was, "It's just not going to work to put representations in and make them think like people." And Searle's view, which I never really found compelling but different, was, "Even if they do, that's not really thinking." So he was basically challenging the operational definition of "quacks like a duck and walks like a duck, it is a duck."

[00:57:55] And he says, "No. If it quacks like thinking and it walks like thinking, it's not thinking because it's not embodied in the same way that human brains are." But he was a much better salesman than Dreyfus and a very combative kind of guy. But they were interesting discussions. As I said, there was not a combative discussion. It was really, let's step back and think about what's working and what's not working. And I think, for me, at that point when we were trying to build mechanisms that would think more like people and not getting very far with it, I said, "Well, maybe we're not getting far because we're barking up the wrong tree."

That's one, of course, of Dreyfus's favorite metaphors, which is, you have somebody who's climbing a tree and says they're getting closer to the moon. Yeah, you're getting closer, but not the way to go. So that was really what started me ruminating about, are we really heading up the right direction? And then there was this completely unexpected additional connection, which is Fernando Flores. And I'll tell you the quick version of his story. Somebody should write a book on him.

[00:58:40] Fernando was a brilliant systems student, sort of like the MSE program here at Stanford, systems in a broad sense of how do they apply to things. And he was in the university, the Catolica, which is the most prestigious university in Chile. The students got involved in politics, and he was part of a student group that was pushing for change. This is now the early 1970s, late 1960s? Late 1960s. Late 1960s because it was during the student uprisings around the world.

There was student unrest in Chile, as everywhere. And in some deal—and this is where I'd love to see the book get written because I don't know the details here—he ended up essentially being provost of the university at the age of 24. I think it was not an official title. But anyway, he was. And then he had been working on the political party of Salvador Allende, who was the democratically elected socialist leader of Chile. And when Allende came into power, he brought Fernando in.

[00:59:58] First, he was vice president of the state-owned industries. Then he was head of the state-owned industries, and then he was minister of economics, minister of finance. And he tells a story about just before the coup, he was being asked to be minister of defense. So it's an amazing trajectory.

Fikes: How did you make contact with him?

Winograd: Unfortunately, the answer is things did not go well in Chile. There was a coup. Allende was killed. Fernando's office in the Palace Moneda was bombed, but he wasn't actually in his office at the time. He ended up in political prison, this island off the south coast of Chile. He was there for three years, and during that time, Amnesty International took on—at that time, I don't know if they still do it this way, but they had prisoners of conscience around the world that they would link up with a chapter, a local chapter, of Amnesty International to pressure. Send letters to the State Department and do all the things that you can imagine trying to pressure the Chilean government to let them out.

He was not a big threat to them. He was a whiz kid. He got to where he was because he was very smart, and he had people who put him in positions. He wasn't the head of the local communist party, who was also in prison with him actually, who had a constituency that could be a problem. He was just this lone brilliant guy. So they were not so reluctant to get him out.

[01:01:40] They gave it to the San Francisco chapter, and the San Francisco chapter of Amnesty International happened to include George Dantzig and Bob Floyd, both of whom were professors at Stanford. So one of the conditions that the Chilean government put on his release was that he had to have a job waiting for him out of the country. They said, "Okay, he can go. But he's got to go, really go. And if he doesn't have a job, he's not really gone." In those days, the funding agencies were a little more flexible about how you would justify hiring somebody. They've gotten more conscious now.

So Floyd and Dantzig had some NSF money, and they said, "Okay, we'll hire you as a research associate." Based on nothing to do with what they were doing in Stanford. It was like, "We have this money, and you need a job. Okay. You're here." So he was literally in prison on this island in Chile, and a day later, was in Palo Alto.

Fikes: What kind of background did he have at that point? He became a very important colleague of yours and co-author of this very, very important book.

Winograd: So when you say background, it's a little tricky with him because he's a Renaissance man, to use the metaphor. I used to say, Fernando is the only person I know who can read four difficult books before breakfast. He reads voraciously. He's interested in everything. In prison, he had visits and talked to a Chilean biologist named Maturana, who was doing theory of biology, who had actually been at MIT earlier. He read philosophy. He read political science. He read management. So his formal background really isn't what's relevant. It's that he was putting together ideas from all these different fields.

[01:03:54] So we got together. I'll cut it shorter; I can tell you the whole story about how, and he said, "It's interesting. When you talk to me about artificial intelligence, I see things in there that resonate with what my experience is in politics and in philosophy. So why don't we do some reading together and discuss it?"

So we started off reading some stuff on philosophy of science—actually, it was the first book we read to discuss—and then gradually worked into a discussion, a discourse of what were the underlying issues, not the technical details, but the underlying issues in artificial intelligence and computing and in human interaction, where he was human-human interaction, in management, for example, and in politics, which were his primary background. And we started writing a paper, and then it got longer. It was going to be a monograph. And George Lakoff, who I mentioned, at Berkeley, was editor of a book series. He said, "Well, you can put it in my book series."

I first met him [Flores] in 1976, which is when he came to Stanford. We actually published that book in 1986, and that was not because we felt it was done but because everybody was saying, "You're too long. You've got to get this out."

Fikes: Did you consider yourself still working on AI during that time?

Winograd: [01:05:33] If you take the period from, say, 1976 when I started that discussion till the early 1980s at some point—and I can't tell you exactly; there was no event that marked the date—I would have said, "Oh, yeah, I'm still in AI. But I'm looking at the philosophical side of it. I'm not doing technical research in AI." And I think, at that point, I would have said, "Sure, I'm not doing technical AI research. I'm in AI but more from let's think about the principles."

Then what happened, moving farther in the future, is—that was fine for me. I could write books and give talks and so on; it didn't work very well for graduate students. You can't say to a graduate student in computer science, "Why don't you think philosophically about AI?" [laughs]

You could say that. I did. But not as your dissertation. I had a course for many years—which Reid Hoffmann actually took and I think that was where he got into this—where I said to graduate students, "Let's think philosophically about AI." But the next step was a dissertation had to go in a different direction.

Fikes: So hang on, that's very interesting. So you're talking to them about going in one direction, but it's not a direction that they could get their PhD. They

have to do something entirely different. So did you help them with the technical side or with what they had to do to get it?

Winograd: If they did technical stuff, then I would help them with it. But I guess to be honest, there was a period sort of not much happening with graduate students. And then they started working on things which didn't really look like AI. They looked like human-computer interaction. And so at some point in the 1980s, late 1980s, I made this conscious decision to say, "I'm not doing AI. What I am doing and what you can do your dissertation in is human-computer interaction." But that was at the end of a pretty long period of just sort of, well, I don't know.

Fikes: [laughs] Right, right. So the book was very important. It's still read and talked about—In *Understanding Computers and Cognition,* you looked at computers as they existed then and set out new directions for future development. So you've talked about your collaboration with Fernando Flores. What were some of your observations about computers as they existed in the 1980s? And what were some of the directions you described for future development**?**

Winograd: [01:08:20] Let me pick out the sort of core issue, which I've been hinting at in the earlier parts, just talking about representation. The core issue was we program computers by essentially thinking about what we would think if we were doing the same thing they're doing. The program is a set of steps. The steps are based on basic primitives that we can define and so on. And the underlying message of the book is, that's not going to get us to human-like thought. It's perfectly useful. Computers, programming, did all sorts of

great useful stuff. But it's not like human thought. And what do you do about that? If you look at that book carefully, it says, there is this whole other direction people do of connections as neural net kind of stuff.

And of course, that had been going on since the 1940s but hadn't gotten very far either. And what we said is, our philosophical background doesn't preclude that working. It might work. It's not working at this point. But we don't know. So there was kind of a hedge there where it says, we don't know. The principles we're talking about don't apply to that because it's not based on having a representation that you articulate and so on. So we can't say the same principles that it won't work.

[01:09:55] But that's not where we're going. I often said, back in those days, I could have either gone to human-computer interaction or gone off and studied real neuroscience. If you want to think like a brain, you have to understand how brains actually think. And neuroscience is trying to understand how brains actually do it, even though I was too far along in my career to make that big a switch, although it's fascinating.

Fikes: So what were the principles?

Winograd: The principles are that humans don't react to the world by having representations of it. They react to the world by having a coupling. And this is a biological theory from Maturana, who I mentioned earlier. Structural coupling is his word, which is something triggers a set of activities in the organism—and he was looking at all sorts of organisms, not just people—which in turn lead to perturbations, to use his term, which in turn lead to behaviors. But that path is not based on representing what they see. To give

a very simple example, imagine a little one-celled animal that the membrane around it gets slightly more firm when it's exposed to light. What's that animal going to do?

[01:11:28] Well, it turns out, if you shine a light on one side of it, that'll get a little firm. It'll squish it out the other side, and it'll move away from the light. So you can say, "Okay, this organism understands moving away from light." And of course, if you look at it, the answer is it doesn't understand anything. It just happens to be built in such a way that will effectively move away from light. With no representation of light, with no logic, no decision making, it just does it. Humans are much more like that than we want to admit. We think, "Oh, we're smart. We would say that's a light and then move way." But most of what we do—except for the conscious linguistic thinking, which we also do—is more like that amoeba. What does that imply for building intelligence? It says if you're going to build intelligence, you have to understand the whole structure of embodiment. And that's part of the theory that came from the biology side. It's nothing more embodied than how your cell moves. And the history, the experience, which is not represented as history but has changed—and again, that's another part of the theory, that what history means is you have been changed by what you've been through.

It doesn't mean you have a representation of what you've been through, but you have been changed by it. And that's the history which determines what direction you're going to move, to put it in simple terms. So it was a rejection of—and this also is tied in with phenomenological

philosophy. Heidegger comes up a lot in the book. And Heidegger was one of the first philosophers to start from the representational logical side and say, "That's not what people are doing."

[01:13:16] The people have what he calls "thrownness." He has all sorts of weird terms. There's a sense in which something happens—again, the amoeba is a very primitive example of thrownness, which is different from reflection. You don't reflect on what you're doing. You're thrown into what you're doing by your own structure. To a large extent, that structure has a biological basis. And this is one of Dreyfus's points, he's a scholar of Heidegger, that a robot that doesn't have the same thrownness as a human body won't respond like a human body. Even though you can get it to climb a wall or whatever it is you're trying to get it to do, but it's not going to be doing it in the way that a human would.

Fikes: So as you began to have that understanding and write about it, what does that mean in terms of the direction of your career?

Winograd: In terms of my career, the first obvious point is I was trying to build AI, and this said, "This is not how you're going to build it. Maybe there's other ways, but this is not how you're going to build it." Well, what did I really want to build an AI for? That's the basic question. Why did I care about that? And there are two very different motivations. One is—let's call it the theory understanding motivation. I want to understand how brains, how people work, and I could use building something as a kind of thinking tool to get there. If I could build something and imitate it, that gives me a good starting point to see how it is really working. That's the part I was giving up.

[01:15:00] The other is if you make computers do things that are appearing intelligent, they're good for doing things. They work well for people. So let's look at that side instead. I'm really at heart an engineer, not a theoretician. So my motivation is more what's going to be useful than what is that underlying theory. Although I like theory, I dabble in it. But at heart, I'm really concerned with pragmatics and how do you get things done. So if you say, how are computers going to be useful for people? One potential answer is make them smart. And another is make them work well with what people do.

Don't make the computer think *like* a person. Make it work well with a thinking person. And that, of course, is the branch off into human-computer interaction. So at some point in the 1980s, I said, okay, I'm really doing human-computer interaction. I'm really not doing artificial intelligence.

Fikes: Where was your office at that point? When did you move out of the AI lab?

Winograd: I had moved out of the AI Lab prior to that, sometime in the 1970s. I'm terrible at dates, but I moved into Polya Hall, which was where the chair had the Computer Science [Department] office, people like Knuth and so on. And I don't remember whether it was my choice or whether the lab shifted—anyway, I was in Polya Hall.

Fikes: That's a significant change, right?

Winograd: Yeah, it was certainly what I said before, now you're on-campus, you're working and teaching courses and students. It may have had to do that, but it's interesting, I really don't remember when or how. I just remember I had

an office in Polya Hall. And then from Polya, we moved to Margaret Jacks Hall. It must have been there when you came.

Fikes: Yes.

Winograd: And I'm sure you could easily find out the date of that, which I don't remember. I believe by the time we were talking about now, I was in Margaret Jacks.

Fikes: Perhaps you can make a comment about this in terms of the history of the department and Stanford. The department was spread out in multiple locations, right?

Winograd: Yes.

Fikes: The extreme one, of course, was McCarthy. But Feigenbaum—

Winograd: Welch Road.

Fikes: —was over off wherever we were. Anyway, was that difficult? Was that a problem? Was that an issue?

Winograd: [01:17:55] When there was a basic view that Computer Science was a collection of people doing interesting research in different directions, and they had their graduate students who would work with them and have offices near wherever they were, then what's the big deal? Why do I need to have an office near somebody who's doing something different? And I've used the word "fiefdom" a couple of times. It was really independent groups sort of doing their thing. And other people either liked or didn't like or tolerated what they were doing. I think AI was tolerated initially, but it came to be more respectable.

There was a whole period during which—and this is while I was still up at the AI Lab in the hills—there was this big debate about computer system building. People like Mark Linton and Brian Reed, who built things. Brian did Scribe, which was a computer program for typesetting, sort of an ancestor of Adobe. I forgot what Mark Linton did, something in hardware. And that was viewed, "Well, that's not really computer science. That's engineering." There was a shift mainly for economic reasons. "Okay, well, that stuff, let's call it computer science even though it really is engineering, not like us math people who are really scientists."

[01:19:27] And AI was always sort of on that cusp. "Yeah, it's sort of AI. But it's speculative. It's not really computer science." Anyway, there is a point at which the university—and I don't know the politics at the high university level—said, "Look, you've got to teach undergraduates. It's nice you have graduate students. It's nice you do research. That's what Stanford's here for after all. But there are always undergraduates who want to take computer science. And you teach some courses, that's fine. But you need a major. You need to actually create an undergraduate major and have a coordinated coherent teaching program, not just every professor gets to teach whatever they happen to like."

And there was a lot of resistance to that. People really did not want to become undergraduate teaching oriented.

Fikes: The research university syndrome, right?

Winograd: Exactly. But somebody was insistent. You really do need to create a major and start to have a real program for teaching. And that was during the period we were in Margaret Jacks that that really gelled.

Fikes: So did you teach undergraduate courses?

Winograd: [01:20:50] Early on, no. So early on, this is, again, exactly what I was saying. I taught courses in my specialty. I had natural language courses. I just ran into a guy in a grocery store parking lot last week who said, "You probably don't remember me. But I took your course back in 1970." And by the date, I could know it was the natural language course. And of course, he was very complimentary. I also taught a beginning programming course. Two interesting facts about it. One is the language we taught for programming was ALGOL W, which was Niklaus Wirth's version of ALGOL. He was at Stanford various points in his history. And the second is students wrote programs on paper, punched them into decks of cards, took the cards to the center over in Pine Hall, put them into the card reader, and waited for their printouts. That was probably in the last few years of non-interactive computing programming for teaching, but when I started teaching that course, it was punch cards.

Fikes: Well, looking at the sweep of history, your early history, we were talking about MIT, et cetera, et cetera, that's the way computing was for so many people. You submit it, you put together punch cards. You put them in for one run, and you get back a result. And then you do it over and over again.

Winograd: And that was the mindset. The punch card mindset was something that I avoided. First of all, as I mentioned, in my beginning job, I had my own

personal computer that I punched the buttons on. So from the beginning, I was doing something interactive. And then when I got to MIT, they had some of the most advanced early interactive systems. Project MAC in AI Lab had—I never did punch cards. Oh, it's not true. I did one course where I programmed musical theory—that was sort of interesting—which was a punch card course.

[01:23:09] But other than that one course, everything was interactive. And everything from there on. Of course, I got to the Stanford AI Lab, and they had the SAIL system, and again, one of the pioneering interactive systems. So in my mind, it was never the punch card deck, except for these odd things like teaching undergraduate programming.

Fikes: I think that's fascinating. It must have a big impact on the arc of your career. So many people involved in computing, myself included, it was punch cards for a very, very long time. And so there was no sense of interacting with this thing.

Winograd: Exactly.

Fikes: You seem to have avoided that from the early days. I think that's very interesting.

Winograd: Interesting thing to happen to fall into. That's where I was.

Fikes: Let's talk about the dawn of the HCI. Eventually, there was an HCI program initiated apparently in 1990. So talk about that and how that was accepted in the department. Or did anybody care?

Winograd: [01:24:26] So let me go back to the history of HCI, just to put context. Up until roughly 1984—and here, I can pin down a very specific date—people

assumed that interacting with computers was the equivalent of punch cards. There were variations. And it was only being done, the only people who needed to do that, were highly skilled scientists, technicians, whatever—data processing people. And then came "the computer for the rest of us." And of course, the work at PARC had anticipated this. But from a social larger historical point of view, all of a sudden in 1984, Steve Jobs was trying to sell computers to architects and students and who knows, everybody who's not a trained computer scientist, who's not a trained physicist who had been happy to write Fortran programs.

[01:25:24] And they said, "Oh my god, how are we going to make it usable for those people?" And of course, they had a big head start from what had been done at PARC with the Alto and all that. So the field that we now think of as human-computer interaction really sprung into being at that point.

There had been people working on things like that before. But it was really—to become a major field, all of a sudden, it has social importance and had commercial importance. Of course, in any research environment, commercial importance means funding. So that's got to be where you go. Funding makes a difference.

So during the early years, the human-computer interaction field was dominated by two things. One was the rough and ready practical stuff—building things that people could use. Engelbart built the mouse, that kind of stuff. And the other was oriented to cognitive psychology. Carnegie Mellon was the home of this—Newell and Simon, human information

processing. There was a whole world. And that also was at PARC. The early days of PARC, they had a group. Allen Newell was really the leader. He wasn't there formally but he was the intellectual leader of this group. Stu Card was his student. Actually, Stu is somebody else you might be interested in interviewing. What happened is the people who were doing the rough and ready stuff said, "Well, we should be more organized, more theoretical. There should be a field. It shouldn't just be a lot of hackers working on a lot of things that are good for using computers." The first conference was in 1984 also—1983 or 1984, it's right around then.

Fikes: So what was HCI at that point?

Winograd: [01:27:27] It was a scattershot of things. So one branch of it—and the one that people always sort of leaned on to say, "See? We're respectable,"—was the cognitive theory side. One of the results that every student learned is Fitts's Law. So Fitts's Law has to do with how much time it takes you to do an action, such as pointing to something on a screen. And it has to do with the distance. The details are not that interesting, but it was real science. You could do experiments. You could do measurements. You could make improvements on the theory. So that was people who wanted to feel respectable to their colleagues in psychology and so on.

The other was the creative invention side. "Oh, well, what if you could have multiple windows? You didn't have to have just one thing on the screen. You could have multiple windows at once. That's a great idea."

So Alan Kay put that into the Dynabook or into the Alto eventually, but his version of that. So you might say Stu Card and Alan Kay, these

people who are at Xerox PARC, are the two poles there. And from a commercial point of view, obviously, the Alan Kay side is what mattered. The more good ideas you could come up with, the more creative you could be about what people could do with these things, the more you'd sell. The field has never been something that has a coherent canon, where you say, "If you want to know HCI, here are exactly the things you need to learn."

[01:29:09] It's been more, "Here's a lot of different approaches people have taken and where they've gotten to with those and how you can apply them to what you're doing." And I think that was why, to pick up on what you said, it was never taken that seriously by the computer scientists. And by that, I mean people who would use the word scientist in their own self-definition. When you say to somebody, "Are you a scientist?," I think many people in computers would say no. In fact, when we started this organization, which we'll come to quite a bit later, on social issues, we said, "Should we call it Computer Scientists for Social Responsibility?" And they said, "No, because a lot of the people we care about in the computing field don't think of themselves as scientists." So it became Computer Professionals for Social Responsibility because they do think of themselves as professionals.

Fikes: Yeah, that's great.

Winograd: And then there was another group called Computer People for Social Responsibility. But then that played on the tropes of the 1960s of "power to the people" and so on. I think most of the people in the Computer Science Department saw it as, "Oh, yeah, there are these guys at PARC. And they're

doing cool stuff." And that was what they had had exposure to, of course, here at Stanford. "But that's not computer science. It's just cool engineering. And if we're going to hire research faculty, we certainly don't want to focus on cool engineering. We want to focus on theory, on theoretical stuff."

[01:30:45] So there's an anecdote. I don't have a date for this. I believe it's when Jeff Ullman was chair, so that could date it to some extent. But we had an offsite retreat for the department, this nice place up in the mountains. One of the exercises was, "We're going to be hiring because, of course, the department's been growing ever since it started. What fields should we hire people in?" And they had a big white board. And they said, "Okay, everybody can call out whatever they think are the fields we should be hiring in. And we'll list them all, then we'll vote." And so, of course, the first thing everybody does is the field closest to them because they want more colleagues. [laughter] But it was the whole department there, so no one group could just do that. They listed "more mathematical theory" or this, that, or the other. And I think there were eleven that got on the board and said, "Okay, let's vote." It wasn't a vote at that point. "Let's discuss." Right off the bat, David Dill, who had been at CMU and was in much more hardcore mathematical algorithm stuff—said, "Well, I think we should take human-computer interaction off the list because I've been at Carnegie, and I see what that is, and it's not worth it."

Fikes: Wow.

Winograd: Now, of course, Carnegie is where human-computer interaction meant applied psychology, and he didn't think applied psychology was right. So it didn't get taken off the list. But it did get voted next to last. So what's interesting about the fact that it was next to last is, again, part of the culture of the department, the one that actually came in last was software engineering.

Fikes: No kidding?

Winograd: And you would initially say, "Wait a minute. That's real computing. That's not weird psychology stuff." But the idea was it wasn't theoretical enough. Software engineering was a set of applied methods, and a set of applied methods wasn't respectable for hiring research faculty. Not that it shouldn't get done; somebody should do it. But it's not what you want for hiring research faculty. It reflected a general culture about what's important as well as the fact that HCI was not in that realm.

[End of Part 1 of 2 – Interview with Terry Winograd – February 14, 2023]

S T A N F O R D U N I V E R S I T Y

PROJECT:	STANFORD FACULTY ORAL HISTORY PROJECT
INTERVIEWEE:	TERRY WINOGRAD
INTERVIEWER:	RICHARD FIKES
DATE OF INTERVIEW:	FEBRUARY 16, 2023
PART:	2 OF 2

Fikes: I'm Richard Fikes, and this is a continuation of the interview of Stanford computer science professor emeritus Terry Winograd. In our first session, we talked about your early life, your time as a student, your work at the Palo Alto Research Center, your early years in the faculty at Stanford, your move away from classical artificial intelligence research, and the beginnings of your focus on human computer interaction. I would like to begin today by getting a broader view of who you are as an adult and to consider influences that were informative for you in addition to the educational ones that we've already discussed. Let's first take a more in-depth look at your early years. Tell us more about your family life growing up and your early schooling.

Winograd: [00:00:50] Glad to do that. I was the oldest of four boys in my family and grew up in a small town in Colorado. And from the beginning, I think my parents said, "Oh, this kid is going to be really genius." Of course, all parents say that, but I certainly had the expectation that I was going to be really good at school, really smart and stuff. And as a result, I think I grew up very much with the confidence that I was going to do well. I didn't have

a sense of, "Oh, can I do it? Can I not?" And being the first, of course, nobody to compare to.

I was reading by the time I was three. I very clearly can actually picture in my mind the day that my mother took me to the kindergarten class at our local school. And they wanted to see if I should skip kindergarten. So they gave me a book, and I read it to the other kids. I don't remember what book it was, but probably Dick and Jane.

Fikes: You didn't do kindergarten, right? They threw you out?

Winograd: They actually took me to the fourth-grade room, and they handed me one of the fourth grader books and said, "Here. Can you read this?" I did a passable job at that, too, so they moved me into first grade. So I skipped kindergarten. Then we moved to a new house, and I went to a different grade school, and I skipped second grade. So I was actually two years younger than the kids in my class. I think that was one of several factors which led me to always not be one of the gang, to be—I wouldn't say a loner. It wasn't like I didn't talk to people, but I always didn't think of myself as part of a group, in that sense.

[00:02:48] We were Jewish in a town that was almost entirely Christian. Different holidays, different customs. So I grew up sort of as an outsider. I mean, all the kids respected me and thought, "Oh, this is the genius in our class," but they didn't play with me.

Fikes: Any problems about being Jewish in that community?

Winograd: Not by the time I was growing up. My grandfather, when he moved there—this is now back in the 1920s—had a cross burned on his lawn. So

there was certainly that undercurrent, but by the 1950s, when I was growing up, it wasn't active. I'm sure if I had been a very subtle analyst of what people were saying to me, I would have said, "Ah, I see in that comment there—", but there was nothing blatant or overt. But it did lead to this sense—and in my family as a whole, not just me—of, "We're not part of the neighborhood."

Fikes: Right.

Winograd: "We're here. We're happy to be here and be friendly with everybody, but we're different." I think that lasted—I mean, it's lasted into, to move it forward, my interactions in the Computer Science Department, that I never felt I needed to get acceptance as one of the gang. But I was there, and I respected what they did. They respected what I did, but I was different. I think that element of it really did make a big difference in my career choices and the things I did later. So I was this sort of nice but distant genius kid in school. When I got to high school, the math teacher said, "Look, you do not need to sit through the geometry class. Why don't I give you a programmed instruction thing on the new math." It was just coming out. The new math was the big new thing.

[00:04:46] So I remember sitting in the geometry class while they were all doing geometry, which I'd already done, doing this programmed text. In those days programmed texts said, "If you think the answer is this, turn to page so-and-so." It wasn't programmed on a computer—learning about set theory. That was what they were trying to teach in the new math for high school level. Then I got to physics, and the physics teacher said,

"You don't need to sit through all these lectures in physics. There's a sort of stray room"—it was up in the tower of the school—"that nobody's using. Why don't you go up there and do a project?" So my physics class in high school was building a computer, which I did during physics time in this garret room up in the high school.

Fikes: By yourself?

Winograd: By myself. I think that's part of that whole pattern of, "I'm going to do something interesting, and I can do it by myself. I'm not part of a group that does it." Especially in that kind of a school. I mean, I was just off the charts for what everybody else was doing.

Fikes: Right, right.

Winograd: It led to that feeling.

Fikes: Well, fascinating. Do you want to talk about that computer for a minute? You brought it in. You have it.

Winograd: Yeah, I brought it in. So I got interested at one point—and my mother said that she remembers when I was seven. Now, I don't remember exactly when, but I was sitting on the toilet, and I made this great mathematical discovery. I came out and told her the discovery. And the discovery was, if you look at the pattern of squares—one, four, nine—the difference between them is the successive odd numbers. Then if you look at the pattern of cubes, the difference between them is—and I won't go into all of the math. It's trivial math, and it's actually something that Newton developed. That was his starting point. It wasn't what he developed, but it's called the calculus of finite differences.

[00:06:44] Basically, it's looking at things like what calculus looks like, derivatives, but for integers, not infinite decimals. And so when I got to needing to do a science project, I said, "I bet I could do a computer that calculates from that." So it's something I sort of had in the back of my mind, for God knows how long.

Fikes: What was your mother's reaction to that? I mean, did she understand enough of it to appreciate it or encourage it?

Winograd: I think she did. I think she did because it was very elementary. It was a discovery for a seven-year-old. It wasn't high math. So let me talk about my parents a minute. My father went to college in engineering. He also was the genius kid in the high school. At one point, they did an IQ test, and they said, "This is the highest score we've had since your father was here." [Fikes laughs]

He was very engineering oriented. He did electrical engineering in college and then in the Navy, but that wasn't his job. But he certainly encouraged that kind of thinking. He had a workshop in our basement, and I used to go there and build things. My mother was very bright, as you might guess, but not as educated. She, when she would have gone to college, became a nurse's aide during the war in Washington, DC. She wasn't overseas or anything, but she was working to help the nation, the war effort. And that's when they met.

[00:08:31] She actually went to college and got a degree when my youngest brother was in college. She said, "I'm going to go back and get my degree." She was on the board of trustees for the local college. She started a

mental health center. She was active in the United Nations efforts. She was a very engaged and active person but not academic. So I think, when she saw that, it was not surprising to her.

Fikes: You had commented in our first session that you wanted to have a career in arithmetic.

Winograd: Right.

Fikes: I wondered what your family's reaction was to that, if they knew about it or what their expectations were for you, for your career and education.

Winograd: Oh, well, of course they thought it was cute. But their expectation was always that I would be in the sciences. I think they wouldn't have said computer science, of course, but I mean, what I loved to do was sit and read science books. So that's what I was doing when I was six, seven, eight, nine years old. You project forward, and you say, "Okay, that's going to be the direction."

Fikes: You did bring in this computer that you built.

Winograd: I built that in order to do the calculation, which is very simple, is, given a series of points—and they were integers because I only had integers here—on a quadratic curve, what is the equation for that quadratic? It's, as I say, a very simple formula that you can do. On my first attempt, I thought, "Well, I could do that with gears. If I could build gears of the right sizes and then have them rotate together—" I actually designed a gear-based machine that did that. Then we tried to do it. Of course, you can't build real gears of arbitrary sizes that you want.

[00:10:28] So we carved them—and I say "we" here. My father helped with some of the hardware issues—the clear plastic, Plexiglass disks that meshed against it. Well, it turns out Plexiglass doesn't really mesh very well because it's smooth. So we put rubber bands around. After a bunch of mechanical engineering, we gave up and decided that was not going to work. I said, "Well I could do it electrically, with resistors and measurements." So what this device is, it used to have four knobs on the top row. It only has three now, and what's behind those is a helical potentiometer. A potentiometer, it's like you have a volume control in a car, in a radio. As you move it, it changes the resistance, which, in turn, changes the sound or whatever changes.

[00:11:26] The helical one went around and around and around and around. So you got a lot more resolution. You entered the data into those four gang switches at the bottom. And that was the values at the four points that you were testing. Then you rotated those knobs until the meter told you had the right answer.

Fikes: [laughs] Okay.

Winograd: So that's how it worked. And it worked. It calculated, and I did a project.

Fikes: So what did you do with it? So you built it. It seemed to work.

Winograd: I built it, and I took it to the science fair. It was done as a science fair project.

Fikes: I see. okay. This was at your school, your high school?

Winograd: Well, there was a local one. It was not just our school, but some local area science fair. Then from there, I took it to the state science fair. The

comments I got on it, which actually are interesting thinking about the history of computer science later, is, "Well, it's really cool, but it's engineering not science."

Fikes: [laughs] Yes.

Winograd: And the guy who won, who was a co-student of mine at my high school, did experiments on—I think it was flatworms or some kind of little animal, where he had multiple little things, and then he tested them and measured them. Very classical experimental lab science. And they said, "That's science."

Fikes: I mean, it's a question that all of us in computer science deal with throughout our careers. But I wonder whether you consider yourself a computer scientist in the work that you did in HCI and then in design, et cetera, et cetera or what?

Winograd: Why are we called computer science? An interesting question. And the science part is a status question. If you're in science, you have high status. Well, it used to be. Nowadays, who knows what's happening, but it used to be science had high status. Engineering was you wore your engineer boots, and you went out, and you built things.

[00:13:42] I think from the beginning, there was a desire on the part of people doing computing in academia to aspire to, "We are one of the sciences. And this is going to be big like physics. It will be the underlying science for what people can do." So I think there's a science envy. Why do we have social science and economic science? I mean, everybody who's got

an academic field wants it to be called a science because that means it's serious and solid and so on.

[00:14:15] Computer science is like having something back in the early part of the previous century, in the late 1800s, called camera science. And who would be in your camera science department? Well, you'd have people doing optics, people doing emulsions of film, people doing aesthetics, etc. because there's this very valuable new thing called a camera, and everybody's trying to come up with different ways to think about it. So we'll call them all camera science.

But of course, people doing optics and people doing aesthetics and people doing chemistry aren't really doing the same thing. They're doing it with the same device, the same camera, but it's really not the same thing. I think computer science is like that. Everybody who is doing something to do with computers said, "Well, we're part of computer science." But it was really very different kinds of intellectual approaches. So doing the mathematics of algorithms was mathematics. It was about computers, but it was mathematics. Whereas building chip circuits was electrical engineering, but it was computers.

[00:15:19] I think from the beginning, computer science has been kind of a catchall. If it has to do with computers and it's interesting and solid, sure, it's computer science. That was resisted. I think the people who started off thinking of it as a mathematical discipline—people, including at Stanford—said, "No the other stuff isn't really computer science. It's useful stuff. It's not really science. It may be engineering. It may be whatever." So

I've always thought of myself as being as much in computer science as anybody because I was doing stuff around computers.

Fikes: You talked last time about the difficulty of getting new faculty in HCI because the department wanted research professors. And at least some of them didn't consider what? Didn't consider HCI an area where research was being done?

Winograd: There's a question about what constitutes academic research. There's two completely different directions you can come at it. One is what the funding agencies will support. If they'll give you a research grant, it must be research. Of course, that is like computer science. It can be anything because they're different funding agencies with very different motivations. But it's driven by needing funding for the research. The other is a sort of model based on the physical sciences. We say, what's research? If you said to one of these AI bots, "Draw a picture of somebody doing research," what would they be wearing? A white coat.

Fikes: A lab coat. Right, right.

Winograd: Why? Because the model we all have about research is standing in a laboratory with a white coat and a bunch of test tubes and so on. And it has certain intellectual basis of, you do multiple experiments, you do statistics, you have hypotheses. I mean, this is classical research for probably centuries. So if you say, we're going to hire professors to do research, there's this question of what constitutes research? If you hire somebody and they go build a cool device, is that research? Or in industry, they talk about R&D, research and development.

Fikes: Right.

Winograd: So I think there was always a question. And human computer interaction has a very different focus. And one way to think of that is, if you say—let's go back to analogies. I put together a group of people to have a discussion about computers and design back in the 1990s. We actually went off for a retreat by the seaside. It was lovely. I got a bunch of people who were doing design in different fields—industrial design, mechanical design, urban design. One of the people was Mitch Kapor, who had created Lotus, the software company that basically got the IBM PC to become popular because it did spreadsheets and documents and stuff.

[00:18:34] He had this analogy, which I really liked—and he's got a chapter in the book—which is, if you want to build a house, who do you call to figure out what house to build? Now, there's one answer, which is the engineering answer because the house would better stand up and not shake in the wind and an earthquake and all the obvious physical things that make a house solid and work. That's engineering.

[00:18:59] But that's not who you start with. You start with an architect. The architect doesn't say to you, "How many pounds of force do you expect to have on the roof?" The architect says, "How do you use your space? Do you want a big, open space like this? Do you want something cozy? Do you want a place where lots of people can gather, or do you want each person in the family to have their own space?" I mean, there's a whole set of questions which have to do with the interaction between the people

and what they're doing and what they like and the space that's being built for them. When you're doing a computer system, it's the same.

[00:19:32] What Kapor said is, "I'm a designer not a programmer." He was not a hacker. He didn't build programs. He told his programmers what to do, but he was thinking at that next level of, "How do you want to use this? And what does it mean to you as you use it? What it makes it useful versus not useful." All of the questions about the human context and interaction with the device rather than the mechanics of what the device does. So from the beginning, human-computer interaction was taking this "What are you doing with it?" as opposed to "How does it run?" kind of view.

When you talk about design, it's really focusing on that aspect. So if you talk about a designer, classically, of course, people talk about a designer, you say, "Oh, you mean somebody who decorates your house or somebody who does graphics." They think about the visual design. Visual design is one part of it and has always been a part of HCI, but it's a larger picture than that. It's a picture not of what does it look like, but how do you think about and how do you promote the way in which it fits with people and what they do?

Fikes: You taught your first course in HCI about 1990, I think. So what did you teach in that course?

Winograd: Okay. So then the question becomes—let's assume that we agree that that skill of being able to make the things you build work well with people, do

the things that they need and want and the way they understand and all of those human sides of it—how do you train somebody to do that?

[00:21:22] The first obvious answer is, it isn't engineering in the classical sense of finding the right formulas and applying them. If I want to know how to keep my roof from falling down, there's a bunch of formulas for how big to make roof beams. But if I want to know how big to make the space for the people to use, there's no formula. It's not that kind of question.

So doing a class where you teach by the textbook and you give people a bunch of crystallized knowledge, to put it in that form, it doesn't work. What does work? Well, one answer is, you work with the people because that's what you need to do and find out what is and isn't working for them. So it's an experimental, incremental kind of process, not a deductive process. You don't sit back and figure out what's going to work. You try it and then find out what doesn't and what does and continue on. You need to be able to experiment. A lot of computer science is like that. If you go to computer science courses in systems—not the mathematical part, but systems—that's what people do. They build things and see how they work.

[00:22:27] The problem with doing that with human-computer interaction is to see how they work involves people. You can't just say, "Oh, there are no bugs." You have to say, "Oh, well, we tried it. And here is what they said and what they did and what they didn't do and what they couldn't do."

You need real experience building things—because that's what you're being trained to do—and trying them out and getting reactions and interactions from people about what they do. What that leads to is that courses have to be centered on building projects, not on learning the textbook. From the beginning, the courses we did at Stanford were project-based courses. Stanford wasn't the only place.

Then you have the fact that when you start talking about what people do, getting the brightest programmer isn't all you need. It's nice to have that, but there's a whole bunch of different kinds of skills. So one of the things that we pushed from the beginning is to try to do interdisciplinary work, not to just have people who were trained in computing technology but people who had backgrounds in psychology or humanities, whatever it is, thinking about people and what they did.

[00:23:43] Kind of not a political but a cultural view that teamwork is going to get you a lot farther than individual work. For certain kinds of things, there are, obviously, lone geniuses but when you try to build something in the real world, you're going to get a lot farther if you have multiple perspectives. Alan Kay, who you know from Xerox PARC and was one of the real founders of the human-computer interaction world, said, "Another point of view is worth a thousand points of IQ." What he meant is, getting multiple points of view on the same problem, you got a lot further than just being really smart and having one point of view that you fixed in.

So the courses that we did—and this led to all sorts of things and basically continued on throughout the years and into the d.school courses—were project-team based. You got together.

Fikes: Let's say you're in day one of the course or something, early in the course, what kind of projects did you encourage the students to do?

Winograd: [00:24:56] What projects to do is, of course, a very interesting question. And let me jump ahead a little to the philosophy that was in the design school, the d.school, which is, there's a series of stages you go through in doing a project. And the first is what they call in a simple formulation, observe. You go out, and you look at the situation you're trying to design for, and you see what's going on, and you talk to people and find out what they think about what they're doing before you even decide what you're going to build. If you're going to try to build something useful for them, you have to understand their context and their uses. Then you can say, "Well, wouldn't it help if you had this? Or would you like to have that?"

So the first step is getting a target user group because, obviously, you can't design everything for everybody and doing investigations with them as best you can of what would actually work, what wouldn't work, why you would do it, how much sense it would make. Now, for a short-term project course, which people are spending something like a quarter of their academic time on for one quarter, if you add up all the hours, it's not a lot of hours. It's hard to do anything in depth for that. I mean, if you're really doing real design, you can spend weeks doing that part of it. But of course, that doesn't make any sense in a quarter course.

[00:26:26] So what we ended up with—and was sort of a compromise—is, they mostly did projects for other students. Because if you say go talk to people who are going to use this—and it's other students, it's easy to do. You go around, you talk.

If you say, well, it's going to be designed for farmers, just getting to the farmers and getting them to talk to you and so on is a whole project. So you couldn't really design for people who were hard to interact with for students. We'll start with the fact that not all of them have cars. How do you get there? So typically, the projects had a lot of, this is going to help you dealing with the cafeteria or with dating or with whatever. It had that sort of—not always. The other one that they liked to do is young kids. And there are kids in the ed school and so on. You can do some research on that. But you have to be honest in what you can do in the amount of time they had for those projects. So you encourage them to take some little piece, not some big, comprehensive thing.

At the d.school, Josh Cohen and I started a course later which did projects in Nairobi, Kenya. So there, there was much more of an attempt to learn from an unfamiliar user group. What we did is we had an initial exploration with some of the students—we couldn't take the whole class over there, but there ended up being some in each team—and we worked with people from the University of Nairobi and with other partners. We were actually initially partly funded by the Nokia Research Center, back when Nokia was a big thing in phones. What was interesting about the Nokia Research Center—it's how we actually got into Kenya because the

guy from Nokia came and gave a talk at Stanford. And we said, "Oh, we were thinking of doing a project somewhere in Africa."

And he said, "Come to Nairobi. That's where our lab is."

It was a social science lab. It was not a technology lab. It wasn't building devices. It was trying to understand in that context—which, of course, is different from the European context—what are the usage and what is the human-computer side of it?

Fikes: Were these undergraduates that you took over there or graduate students?

Winograd: Let me go up on that and then come back down. The answer is graduate students. From the beginning, the d.school had this attitude which is we are not a design school in the sense that people come here at a young age learn to be designers and then leave. That's the old notion of design. You can do that for graphic design. But if you want design in this larger sense, you want to have people who have depth in some field as well as an understanding of the design process.

[00:29:37] So we always talked about T-shaped people. You'll find a lot of diagrams that look like that if you go on the website. A T-shaped person is somebody who has depth in some field—whatever it is, whether it was computing or art, whatever—and has this crossbar, which is what the d.school was doing, which is working together with other disciplines to see how they could interact to create new designs. So we didn't want to take people who didn't have depth already. Otherwise, you get simple toys. If you could get somebody who really was skilled at computing and somebody else who was really skilled at building mechanical objects—and by really

skilled, I mean an undergraduate degree. I don't mean super skilled. But it's a fair amount of depth by the time you get through an undergraduate degree.

[00:30:25] From the beginning, the d.school said, "We are here to take graduate students from multiple departments, not from our own department"—it was not its own department, and it still isn't a department— "and get them a second layer of learning, which has to do with the integration across disciplines and across methods." And so the methods of how do you go about doing this research and then trying things out, building them and so on, was really the focus, not any one of the underlying disciplines.

Fikes: Specifically, the Nairobi project—so you took a group of graduate students over there. Did they stay for the quarter?

Winograd: No. We had a short period, a week or two. We had to fit it into the vacation schedule—finals, vacation, et cetera.

[00:31:18] Then they would come back and work with the team that was then built in the course here and tell an interaction to some extent—it wasn't really able to be really great—but with the students that we were working with there because there were students at the university there who were not members of the project but sort of consultants to the project.

Fikes: What did they build? What was an example of what they built?

Winograd: One of them actually got carried forward after the course was over into a real project that they developed. It was water. It turns out if you're in the slums of Nairobi, there's no city-supplied running water, as you might

guess. There is no, even, informal running water, mostly. What you have is places scattered through the area where somebody trucks in—or pipes in if they have the pipes—water. Then you go, and you carry your cans there, and you fill them up, and you take them home. And that's the water you use.

[00:32:34] It's not like when you were in the country before you came to the city, you went to the well. In the city, it's not a well. It's somebody. As part of the research, [they found that people] used to spend most of a day going around, trying to find out where they had water today. So they said, "Well, look. The technology side of this for almost everything is mobile phones." Because people, even in those kinds of underprivileged situations, tended to have flip phones, not fancy phones, but they all had mobile phones.

Kenya turned out to be one of the real-world leaders in getting mobile payments from the beginning. They had something called M-PESA, which was a way you could transfer money on your mobile phone because they didn't have a highly developed banking system. It was within the cultural background there that if you could do something on your phone, it made sense. So they built a tool for helping people find water. They had three aspects. One is, who has water? Because obviously that's necessary. Two, what are the prices? Because these were private entrepreneurs who are selling this water.

[00:33:48] And, to some extent—we never really got this one working—what's the quality of the water? So if you could go in and test it

and say, this one has fewer germs, that'd be great. But the mechanics of doing that testing [were difficult]. Again, that's part of what you learn. If you were to sit down in the lab and say, "What should we do?," you would say, "Oh, water quality. That's number one." Well, that's great, but if you get out there and you say, "Okay, we need to build these things or put in these meters," or whatever it is, well, it doesn't really work that way. So they actually built the system and had it running. I don't know what happened to it. I mean, it was sort of sputtering along. I think what happened is that one of the students went there full time after finishing Stanford. I think when that student stopped doing it, it sort of drifted off.

[00:34:45] We always said that the main purpose of this course is not that we're going to build great technologies because you can't do that in a one quarter course or even two quarters. It's an educational thing. It's for the students to learn what the real world is like. And one of the things they learned there—in many projects and this one as well—is you need to deal with the people side and who's trying to get money. There's a lot more to it than what it sounds like.

I'll give you one other one. Again, this was in the slum there. The roads are rutted hillsides. It's not like what you think of as a city. People worked in the city. They were probably house cleaners, janitors, whatever. I mean, low level work in the city and a lot of them women.

They had to get from their home to the city in the morning. Once they got to the road, there were these various forms of transportation that they could take, public transportation. But just walking from their house to

the road and back in the evening was dangerous. They could get picked on. They could get robbed. They get raped, whatever. I mean, think about walking around alone in a bad neighborhood.

This idea, which also actually got fully developed and taken up by the local government there, was to have an organized way for groups to walk together so that you wouldn't be walking alone. Again, here comes the cell phone. You can coordinate using cell phone. And you go, you had somebody who you'd call. They called—what is it? Not the guide. They had another word for a paid person who would escort the group. Think of it like a bus route. This person would go from fixed places, where people could meet them. Then they would walk as a group to the place.

Fikes: This is fascinating. So you're taking a sometimes quite broad look, like your example there to determine problems in the society, just like that, the problem of walking alone in the mornings and the evening or the problem of getting water. So this starts off with a really broad observation of a population or of a set of activities. Am I on the right track there?

Winograd: [00:37:47] Let me fill in one other piece there, which is, how do you go about observing something like that? If you just take an American university student and plop them down in that situation, they've got no hope of finding anything. So we worked with local groups. This was in addition to the university groups. These were local nonprofits. It wasn't Bill Gates or something. It was the local groups that worked in these neighborhoods. And we went to them and said, "Can you help us?" So, to some extent, it was the leaders or the people in those groups who guided what to look at.

They certainly are the ones who made any connections to talk to people. I mean, a white foreigner couldn't just walk out into the neighborhood and say, "Could you tell me what's going on?" But if you could say to them, "Okay, can you come to the office of this group you know. We're meeting with them and other people and so on," then we could connect. So it wasn't an undirected "think about anything." It was really starting with guidance from these groups that were already doing something related.

Fikes: So was that generally a part of what you taught in the classes, that you identify a group--and maybe you go and talk to students and say, "What are your problems?"

Winograd: No. They didn't talk to students. This is, actually, an interesting feature, which is we worked with the local students. It was great because, for example, they could translate. They knew English, and they also knew the local languages. We got comments saying, "This is amazing. I've never actually gone into the slum and seen what goes on." Because the students are upper-middle class. It's a totally different class group that you were designing for. So you had to talk to the people who actually lived in a slum, not to the students who went there with you. You went with those students. That was great. But they don't have the insights.

Fikes: And going back to how does the class start, so would you direct them to populations that they might design something for?

Winograd: [00:40:13] The way it started is, in the context of the larger d.school—this is after the d.school had been going for a few years—we said, "Wouldn't it be a great addition to have some projects which looked outside of this

country, outside of the middle class?" I mean, the d.school in some sense, started with IDEO, which is a great company but what it does is products for the American market, I mean, mostly.

There was a medical student that we worked with who was a very ambitious guy. I mean, ambitious in a good sense, very out there, looking for things. And he had been working with a medical aid group in Rwanda. And he said, "Well, you could come and work with the group that I'm working with there."

[00:41:18] We said, "That's an interesting idea," and we started exploring that. Then we had this talk that I mentioned by the research guy from Nokia. And he said, "You don't want to do that. There's not enough intellectual support to do it there. And we already have a lot of background here in Kenya. That'll be much easier to make it happen and make it work." Plus, as I said, Nokia was willing to help support it some. So that was really the motivation to pick that place and that group. He was a Finn named Jussi Impiö. He was not African himself.

He said, "You don't want to go on your own. You want to work with people at the university there. It's not like you're going off into the hinterlands. I mean, Nairobi is a big, modern city with a lot of universities. And we, at Nokia, have already developed some partnerships with some of the faculty at University of Nairobi. So let me introduce you to the faculty there."

[00:42:37] We ended up having two people, Peter Waiganjo Wagacha—I won't try to pronounce their last names—and Dan Orwa was

the other one who we worked with for the whole time we did this project, for about five years. And they were great, both in terms of giving insights to us and in terms of helping connect to the students there.

Fikes: In terms of trying to get an overall picture of your work, were HCI and the d.school part and parcel of the same thing?

Winograd: [00:43:16] There is a pretty linear history there, which is, as I said, the HCI courses, from the beginning, were group projects—they were design-oriented in the sense of, you come up with ideas, you build them, you try them out. That was the nature of the course. And we did those for a number of years. Bill Verplank, who's a designer by training, a mechanical designer originally. He was in the ME Department at Stanford and he and I and a couple of other people did courses of that sort. Then there was an opportunity triggered by a project going on at Apple. I think I talked about this last time. Anyway, they called it their university something program. And they were trying to get more universities to think about human-computer interaction, being Apple, of course.

Joy Mountford is the name of the person who directed this program, and she really pushed it. She was from, again, a design background. So they said, we will provide machines and some financial support and mentoring for any university that wants to do an interdisciplinary design course. Then at the end of the year, everybody gets to come to the Apple headquarters and present their projects to the others and so on. It was a very nice way to try to integrate this. I had had some interactions—I had done a course previously with David Kelley in the Mechanical Engineering Department,

who was the founder of IDEO Design, one of the founders. And we had done an interdisciplinary design course.

[00:45:14] There's an article we wrote about it. So it was a natural fit. We used that to further support and develop it for a couple of years. It was mechanical engineering. The two faculty, the two people teaching it were David and me, and we had students from a variety of fields. We encouraged students from different disciplines. It included some undergraduates, too. It was not just graduates, like the d.school later was. And he had been doing interdisciplinary design things in a variety of ways**.** I don't want to tell the whole d.school funding origin story. I will if you want, but it's not CS.

Fikes: Yep.

Winograd: It was creating this interdisciplinary design program centered in Mechanical Engineering because that's something they had started back in the 1960s or the 1950s even, called the Design Division. So it was a semi-independent part of the Mechanical Engineering Department. And just as we talked about in computer science, there was definitely a feeling among the traditional mechanical engineers that, "This isn't engineering. This should not be in our department. It's not respectable." But there were very smart people doing it, including some of the early founders of that department, and they were able to just maintain their independence and have their own teaching programs and so on.

Fikes: Well, that seems to be a part of your style, right? You were in the Computer Science Department throughout your career, right? But much of the work that you're talking about is across those departmental lines and all

this interdisciplinary work. So did you feel Stanford supported that kind of work? Did you have problems with departmental boundaries**?**

Winograd: [00:47:29] Culturally, Stanford was great about it. I used to compare it to MIT because I was at MIT, obviously, before I came to Stanford. And the sense of interdisciplinary was just missing there. I mean, it was a set of separate towers, stovepipes, whatever you want to call it, and you did not cross boundaries. I mean, I used to describe—when I was a beginning graduate student, I'd go to a party for graduate students, and I would say, "Oh, what are you doing?" They'd say, "Oh, I'm in Linguistics. And what do you do?" "Oh, I'm in the AI Lab." They would turn around and walk away. I mean, it's like, "We don't want to talk to those guys."

Fikes: Wow.

Winograd: So MIT has a very non-interdisciplinary feel. I should be careful because we're talking 50 years ago. And Stanford really, from when I first got there, had that. If you look at the CS department, for example, Ed Feigenbaum's project—I'm blocking on the name there.

Fikes: Knowledge Systems Lab.

Winograd: The Knowledge Systems Lab, from the beginning, had a huge amount of interdisciplinary work with the Medical School.

Fikes: Yes.

Winograd: And that was just taken for granted. Of course, we want to build systems that are experts in something. You've got to have somebody who's in something. You can't just build systems that are experts out of nowhere.

And so we'll collaborate with the doctors. I think that the university as a whole was pretty open to that, encouraged it.

Fikes: And that seems essential to your work; it's what you've been talking about all the way through.

Winograd: Yes.

Fikes: It's interesting that you describe your growing up, working alone in school and over and over again. Then in your career, you've focused on teams and the importance of collaboration.

Winograd: It is interesting—not the sense of being with a group but the sense of finding a partner and a team. And from the beginning of the HCI program, I almost never taught a course that wasn't co-taught. I had Bill Verplank, as I mentioned, early on, but just a whole range of people with different backgrounds and different things. Let me jump into this because it's going to be relevant—the computers ethics course. So I said, one thing that students should be thinking more about is the ethical responsibilities they have if they're going to go out and build these things. And so I got together with Helen Nissenbaum, who was teaching in Philosophy, long time at NYU—now at Cornell Tech. She's moved. And so we developed a course together back in the 1980s on computers, ethics, and social responsibility. What do you think from a philosophical point of view, and what do you know from a technical point of view were the two sides of the equation of thinking about that.

[00:50:32] It was a very successful course from our point of view. We never had a lot of students. One of the things I used to say was, "I don't

want to make this a requirement because then we'll get a bunch of students who don't want to be there." We could teach it with students who wanted to be there, and it made a big difference. So we taught that for a number of years.

Fikes: [0:50:56] You were one of the founders of Computer Professionals for Social Responsibility. You were on the board. You were president of it. You led it for ten or twelve years. So what motivated you to do that? And was that associated with the course of ethics that you were just talking about?

Winograd: My engagement in CPSR was, again, traceable back to my childhood, if you want to say it that way. My parents were always very socially engaged. It was a fairly small, as I said, conservative town that I grew up in. My mother used to say she was one of the six Democrats in the county. But I remember my first political activism was going around with her—and she said she pulled me in the wagon—campaigning for Adlai Stevenson.

Fikes: Oh, wow.

Winograd: Now, people who are old enough will remember that Adlai was the unsuccessful candidate against Eisenhower back in 1952.

Fikes: Right. Twice, wasn't it?

Winograd: Yeah. So I was under ten years old. And they were very active in UN. I remember we used to go out trick or treating for the UN. Again, most younger people don't know about this, but you would go around and ask for coins to be donated to the UN instead of candy. And so, from the beginning, I had a—let me say a globalist view. That word has other connotations, but a view that it was important, as a person, to be thinking

about the world as a whole and trying to help things, not just your own little local pool. I think that was, in some sense, forward.

[00:52:41] You mentioned earlier, what are the consequences of being Jewish? I think one was people who had a church, a local church, really invested a lot of their emotional and that kind of feelings into their church and their community in the church. And the Jewish community didn't have that kind of thing. It was very small and scattered. So I think there was more of a sense of, okay, we're not going to just do our own. We're going to look outward at the larger picture.

So I was always—I mean, we have a photo—if you were collecting photos, I would give you some of these—of me in a march at the time of the Selma march in Alabama. Now, I was in Colorado Springs in college. I wasn't in Selma. But we had a march about supporting the march in Selma. So when I was in college, I was engaged in these kinds of activities. Then the 1960s came. And, of course, student activism was the mode of the 1960s. So when I was in Boston in graduate school, in Cambridge, I was active in a lot of groups that were thinking about—ending the war was their primary motivation, but it was a broader picture.

[00:53:59] Carol was involved—my wife—in a number of medical activism groups. I was one of the founding members of the Boston chapter of something called Computer People for Peace, which was started in Brooklyn. And it was basically, how can we, who are somehow engaged with computers in different ways, try to end this war? I remember one of our big demonstrations we had was at the Honeywell Company, which, in

those days, was a big computer company. You don't think of it anymore, but Honeywell Computers were a major part of the market.

[00:54:40] They also made cluster munitions. And so we had a demonstration basically saying, "Don't make cluster munitions." I mean, they're worse than just any weapon. They're specifically very anti-people kind of weapons. So I had been engaged in those kinds of activities—marching in the streets with the big student things and stuff—when I was a grad student. Then what happened in the late 1970s is a growing sense here—I think there wasn't so much when I got here. Of course, there were big student uprisings in Stanford in the 1960s, in that same time, one of which is why Stanford got rid of SRI.

Fikes: Right, yeah.

Winograd: Anyways, interesting. David Harris, who was one of the leaders of that, just died just last week. So, there wasn't a lot of activism at that point, in the 1970s. It had sort of quieted. The Vietnam War was over. The 1960s students had gone out to get jobs. I mean, a lot of things had moved along, but there were still, of course, the infrastructure part of that. So there was a growing question of how were computers being used to support warfare? There were two different projects which I was connected with, one at Stanford, one at PARC.

[00:56:15] So the one at Stanford, which had started a few years before that—and Bernie Roth, who's academic head of the d.school now—was the faculty member involved, and a bunch of students, on looking at the way in which what Stanford researchers said they were doing when they

talked to the research world was different from what they said when they talked to the military funding agencies.

So they would get a proposal from the research side that said, "We're interested in doing the physics of how light affects various whatever electronic flows and so on." When they put in an application to ARPA, they said, "We want to do night vision for soldiers." And so they put out a booklet of, "Here are side by side what they say they're doing when they talk to us and what they say they're doing when they talk to the military funders."

Fikes: Wow, okay.

Winograd: And the underlying push was, we shouldn't take military funding for specifically military activities. So that was there. Then at PARC, there were the same kind of questions. PARC knew Xerox was not doing anything overtly military, but there were a lot of companies in the industry that were doing things for supporting military. And it came to a head with what was called the Strategic Defense Initiative, which was the idea you would set up a system which could shoot down any missiles that came in and, therefore, you didn't have to worry about nuclear attack.

[00:58:03] Two obvious points. One point is, if you had that, then you could attack because you wouldn't need to worry about being attacked back. So it really lowered the threshold on, would you attack with nuclear missiles? And the other is, it was crazy. Anybody who built a large computer system—and if you say, this is going to be the largest computer system ever built, involving all sorts of complex interactions with rockets and radars and

this, that, and the other, you only get one chance. The first time you really can test it, if it doesn't work, we're dead. That's not how computing works. It's going to have bugs. It's not going to work the first time. Well, there is no second time.

So we got a guy named David Parnas, who was a respected—within the Defense Department, respected computer researcher to testify to the government that this was not a real scheme. It was a publicity scheme, which is what it was.

[00:58:58] And you could argue, from some point of view, that it helped deter Russians from doing whatever they were going to do. But certainly, as a real technical scheme, it was just crazy. It really wasn't going to work. So there was a discussion group at PARC—Brian Smith, Severo Ornstein, Laura Gould, who else? Lucy Suchman. And I was part of that. At one point, we said, why don't we get together? So we had a meeting at Stanford, where the group from PARC or some people from there and the student group and Bernie said, "Let's talk about whether we can do something together." The outcome of that was forming an organization, which was Computer Professionals for Social Responsibility.

Fikes: Fascinating. So did that organization have much impact or what did it do?

Winograd: So an interesting question is how you measure impact? I think there was certainly local impact—local I mean by time, not one place. So having a testimony before Congress, what does it do? Well, you can't point and say, "Therefore, they didn't do this bill." But it certainly shifts the discussion. I think we were able to shift the discussion on that. I think it added to the

weight. This was the era during which if you said, "There's an organization and the last two words of its name are 'social responsibility.' What do they do?" The answer is, "They fight nuclear war."

[01:00:36] So it started with Physicians for Social Responsibility, Helen Caldicott in Australia, actually. By the time we started up, there were Engineers for Social Responsibility, Architects and Planners for Social Responsibility, Psychologists for Social Responsibility. Everybody was saying, "Look, if we plot the curve of where the government is going now, they're going to start a nuclear war eventually. How do we slow that down? How do we get in the way of that? And how do we use our professional status as a means?" Of course, individual citizens can do what individual citizens can do, but if you say, "The body of people in this field say that this is a bad direction to go and here's why—." And as computer scientists, we had special expertise. I mean, we were more at the heart of the system than Landscapers for Social Responsibility.

Fikes: Right.

Winograd: Psychologists, of course, interesting question because it is psychological, to what leads people to go ahead and take actions like starting a war. So I think we were part of a cultural shift to people saying, "Yes, we need to figure out how to stop this." Now, we were not the major leaders in that. I mean, I would say, no question the doctors were. You may have even seen the poster. We have a needlepoint that my mother made of it, which is a beautiful picture of a flower, which says, "Nuclear war is not good for children and other living things." That was a motto from the physicians.

The physicians said, "Nuclear war is not healthy." And you say, "Yeah." [laughter]

Fikes: Right, right.

Winograd: So anyway, it was part of that general cultural shift. It led, I think, a lot of people to think about things that they then took off in different directions. What happened to CPSR is that it didn't continue on as an umbrella organization, but various pieces of what we were dealing with—because there was more to social responsibility than just nuclear war—took off in their own direction.

[01:02:41] The obvious big one is privacy. So from the beginning, one of the issues we were discussing is computers violating privacy, something called the National Crime Information System, which had been in the 1960s, must be in the 1970s, but it was integrating records. The details, I don't even remember. Then as more and more computers got engaged and more databases—in those days it was databases, not networks—there became a lot of issues about, "What do you need to do to keep the computer systems from supporting violations of privacy?" Computer systems don't violate it themselves. It's what you do with them.

And in the end, one of our executive directors from CPSR for a time, Marc Rotenberg, founded a new organization called EPIC, Electronic Privacy Information Center, in Washington, DC. And that has gone on for many, many years and was very influential in privacy legislation and things like that.

Fikes: [01:03:44] What were the day-to-day activities of that group?

Winograd: It was never day to day because these were all professionals who had a full-time job doing something else and were volunteering to be part of it. So it did have periodic things. We had conferences, for example, where we would bring together speakers and write a proceedings, the obvious thing to do at a conference. We had occasional specific things we wanted to pressure somebody to do. We would get people to send messages and write letters and so on, usually to government and Congress, but sometimes it was getting a company to stop doing something they were doing.

[01:04:31] We had white papers. Again, you can do research on a sort of not day-to-day, but just ongoing basis to analyze what was—because again, our tack on this was, we have expertise. If you want to analyze what are the actual threats to privacy from some database, then you can write a paper on that, and then you pass it along to the people who have the influence. There were regular meetings. And they were, I think, community building, to use the right phrase. They weren't like, "Okay, today at this meeting, we have to decide whether to do x or y." It was like, "Okay, let's talk about what the issues are, and who cares about what, and what we're doing."

Fikes: Were these open invitations out to the community?

Winograd: Yeah. I mean, anybody who wanted could join. There were dues, minor dues. I forgot what they were, but they were not a big amount of money. You could become a member and be on the mailing list. By that time, there were emails, so mailing lists could be email.

Fikes: Did Stanford, as a university, support this or was it completely separate from Stanford?

Winograd: Separate. After that very initial meeting, I don't even think we met at Stanford. I may have arranged a meeting room. I don't remember, but it was definitely not an official activity by either the department or the university.

Fikes: [01:06:04] For you personally, after you became emeritus, I presume you had more time that you could spend. Have you continued to be active in social responsibility?

Winograd: I've continued to be active politically, but not in my role as a computer guy. If you say, what are the political, social activities I'm engaged in? I'm on the National Board of a group called Corporate Accountability, which is looking at the ways in which large corporations are messing people up—I mean, you could say it more nicely—and seeing how to fight that. The big one right now is, of course, fossil fuels. It's now becoming more obvious that the fossil fuel companies really knew they were screwing up the climate but didn't admit it. They suppressed the research that they had themselves saying, "If we keep burning this stuff, it's going to mess up the climate." The UN has a meeting every two years, I believe, it's called the COP, C-O-P, the committee of whatever, participants. The chair of it for this year is from one of the fossil fuel companies. This organization is fighting that and saying, "You don't put the fox in the hen house." I mean, they have an ax to grind, which is not, "Save the climate."

Fikes: Right, right.

Winograd: [01:07:39] It's also dealing with tobacco, which it did way back when tobacco was a very different industry. And they again were involved with pushing that the tobacco companies knew it killed people and were hiding that. There's also water privatization, a lot of places where the water systems are getting privatized and are not good for the people there. I don't want to get into details. Then the fourth is on food and the big sugar sellers, and things like that, Pepsi and Coke.

So that's not computer related. In particular, they have not taken on what are the computer corporations doing. I would be more conflicted on that since I'm involved with those, but they've chosen these other areas, which are, historically, much more serious.

[01:08:36] I was on the National Board, until my term ran out last month, of something called Bend the Arc. And it's a Jewish social justice organization which draws its name from a famous quote from Martin Luther King, which, in turn, he picked from a minister in the century before him, which is, "The moral arc of the universe is long, but it bends towards justice." And the statement is, "It doesn't bend itself. People need to get out there and bend it toward justice." So it deals with a lot of policies about race, about incarceration, about a variety of issues. Even though it's a Jewish organization, it does not deal with Israel-Palestine very explicitly and consciously because that gets you into a whole other set of issues.

I've also been a pseudo board member—and I say this because I am much more than just tech support for my wife. So I am tech support, but I also read her all her email and answer a lot of them. Anything she's engaged

in deeply, I'm engaged in with her. She is on the national board of something called J Street, which is involved with the Israel-Palestine issue. So there's a lot of activism and the same social ethical concerns but not directed toward computer questions.

Fikes: Yeah, okay.

Winograd: [01:10:10] I mentioned the fact that Corporate Accountability hadn't taken on tech firms. I mean, what's going on today in the tech world, there's a lot to be concerned about. I don't know if you've seen the film *The Social Dilemma*. This is a film about what companies, like Facebook in particular, do to the psyche of the country and of kids, in particular. I was actually at a conference they had that they filmed for this. So if you see the movie, you'll see me as it pans through the audience. But it was really put together primarily by a former student of mine, Tristan Harris, who's head of something called the Center for Humane Technology in San Francisco. And this is their area of concern. So I'm indirectly connected to that, not part of it actively.

Fikes: [01:11:08] Let's talk about your students for a moment. You've had some very famous students, but I think the initial question I would have is, what do you think attracted students to you, that wanted you to be their advisor? And how did that go?

Winograd: I think among the students I've had, there was a kind of common thread, which is, I didn't have a big research agenda which said, "Here is what we're going to do. You take that part." I really said, "What do you want to do? I'll support you." So it attracted the more independent people, people who

really had an idea what they wanted to do as opposed to somebody who said, "Hey, I'm in graduate school now. What should I do?"

[01:11:54] My method of working with them was always as a kind of mentor and, literally, advisor rather than boss. I didn't tell them what to do. I said, "What problems are you running into? And how can I help figure them out? Have you thought about this? Here is some background." It was a relationship which really gave them a lot of autonomy. And sometimes, I'll get a note and say, "Oh, you've done research—."'To pick one—eye tracking. One student I had, whose name was Manu Kumar, was really interested in controlling the interaction with a computer using people's eye gaze.

The problem with eye gaze is that it's semi-voluntary. I choose what I look at to some extent, but to some extent, my eyes go [waves hand around] because I'm doing things. How do you take that into account in building a device which actually lets you use it? And, of course, the obvious audience for this is people with disabilities, who can't use a mouse or type or point or whatever. I had never done anything on that; I had no real expertise, but I said, "Sure. If you want to do that, I'll advise you. I can give you generic advice on interaction and on those aspects of it." So he went ahead and did it and published a paper. And as I say, I was on the paper. So people say, "Oh, you must do eye gaze."

The students who did the best were ones who really had a feeling for what they wanted to do.

Fikes: [01:13:30] You have two famous students—well, many famous students, actually, from a Silicon Valley point of view, but Larry Page and Sergey Brin. What were they like as students? What do you have to say about that?

Winograd: So Larry and Sergey—people ask me, what were they like? I've had this question hundreds of times.

Fikes: Right.

Winograd: The first answer is, they were smart, but not that much smarter than other Stanford graduate students. The threshold to become a CS grad student at Stanford is so high that just being smart doesn't get you anything. It's the entry requirement.

There were two aspects where they stood out more. One was drive. They knew they wanted to get something done, and they were going to do it. And the other is a kind of open-ended creativity. I mean, Larry would come in, and we'd talk about space tethers. Well, you know this if you follow what the two of them have done with Google since. Everybody's complaining, "Well, they wasted their money on flying cars and this and that and the other." That's how they are. That's what they love. And although that's not the part that made them the money, I think it led them to do what they were doing in a way which said, "Well, why not?" And, of course, the most obvious one is, they had developed this way of finding links, back links.

[01:14:57] To be technical for one minute, the thing that made the web jump ahead of all the other hypertext—because it is what's called hypertext things before it—is everything before it had said, "If I have to

place something in my document that links to you, you should be able to see a link back to me." Two-way linking, which is great. You can do all sorts of wonderful things. The problem is, it requires cooperation and collaboration and all sorts of things, which, in the real world, didn't work—back to human-computer interaction. And what the web said, what the HTML protocol said is, "Don't worry about backlinks. I'll link to you. You don't even know it. That's okay."

I think that twist, that jump of how you think about it that Tim Berners-Lee did really opened it up because then the technology became doable. And all of these fancy hypertext systems that existed before that disappeared, and then came the web. The problem is, you didn't know. If I linked to you, you had no idea that I had linked to you. So what they said is, "How about finding backlinks? How about being able to search the web, find the link from me to you, and then put it in a catalog so you could look at it and say, 'Oh, there's a link to me from here,' or anybody else could look at it and say, 'There's a link to you from here.'"

[01:16:16] They had come up with this idea of doing backlinks. Their original program, they called Backrub. It was sort of a pun, but it was based on backlinks. And they said, "Well, why don't we just do the whole web?"

I said, "That's crazy. There's too much stuff out there. It changes too fast. There's no way you could actually do the whole web that way."

And they said, "Well, let's try."

I think that's what I mean by having both the drive and the creativity. I mean, other people would say, "Yeah, you're right. That's hard. Let's just

pick a good bibliographic database and do it there." And they said, "No. We ought to try it for the whole web."

I don't need to tell the legends of the founding, but they got a whole bunch of machines, and they got a couple of good programmers to work with them, and they took off doing the whole web. There were problems. One big problem was people who they were searching didn't know what this was about. We had to deal with the Stanford Legal Department. We got threatening letters from people saying, "You're stealing our material."

And we'd say, "No, no. We're not stealing it. We're just finding the backlinks." They didn't understand what the hell that meant.

Fikes: So you say "we."

Winograd: We was the group of advisors. So in this case, this legal one, it was basically Hector Garcia-Molina.

Fikes: You were actively involved in that? And you were targeted by the legal department or at least they came to you as well as the students, right?

Winograd: [01:17:43] Right. I mean, just to give one step of context back, Hector and I and Jeff Ulman and maybe other people—I've forgotten exactly—were part of a large digital libraries project that was started back a couple of years before that, funded by a group of federal agencies, including DARPA but also including NSF and somebody else. And it was basically understanding that connectivity was coming. There were all these interesting databases on the web, and they were all stovepipes. They were all separate. And could you integrate them into a, quote, "digital library?"

That was the project that we were working on as a faculty group, in a classical research project sense. In those days, when you talked about information on the internet, it was dedicated databases. There was Lexis and Nexus and a lot of library bibliographic databases. So when you said, "Is there information on the web?," it was these kinds of things. Then the question is, "How do you integrate them into a unified system?" And that's really what our project was doing.

[01:18:53] What we didn't see when we set up that project is that information on the web was no longer going to be distinct chunks like that. It was going to be everybody's webpage; it was going to be spread out in that way. And that was what Larry and Sergey saw and said, "We want to move that way. We want to say, yes, we want a digital library, and the information that we want is everything that anybody is sticking on to the web, regardless." Which raises all sorts of problems like, "What's junk?," which we still face. So that was the jump that they made.

Fikes: Right. So did they take your courses or any of your HCI course or d.school course?

Winograd: I know Larry did. I don't know if Sergey did. I was not Sergey's advisor. Jeff Ullman was Sergey's advisor. But I know Larry took an HCI course. I don't remember exactly which one and when, but it was certainly not a major part of his coursework. Well, he didn't do a lot of coursework anyway. As his advisor, I had to deal with that in terms of departmental requirements. [laughter]

Fikes: I wanted to relate to what you just described, what their observation and what their direction was, relate that to what you were teaching in your courses. Was that or would that have been a legitimate project in one of your courses, to look at the web?

Winograd: I think the project would have been legitimate, but it was too big for a course. If he had said, "I want to do this for a course," I would have said, "No. That's crazy. You can't do that in a course. Do it, but don't do it in a course." But in terms of the style and substance of the project it could have been. It's always hard to know how to claim credit for somebody's attitudes because people develop their attitudes in lots of different ways. But I can remember when it became Google. I can remember going to meetings when I was on sabbatical at Google. And he was just insistent about things like user responsiveness of the home page. I mean, there were all sorts of technical things underneath, but he said, "You can't click and then wait. You have to do whatever you need to do, technically, so that somebody gets a response immediately." Now, that's a key interaction design principle.

Fikes: That's good stuff. You took two different leaves of absence. You went to Interval Research for I don't know how long. I guess a year, and then you consulted, and then you were at Google for another one. Tell us about going to Interval or was that important?

Winograd: [01:21:44] So Interval and Google were two very different kinds of years. It was interesting for me. Interval was started up by David Liddle, with the financial support of Paul Allen, who was co-founder of Microsoft, had plenty of money. And, in some sense, it was, "We want to be the next

Xerox PARC." If you go back and you look at Xerox PARC, they were building things that weren't currently in the market, that people didn't currently see as what needed to be used and then became everybody later on. Xerox didn't do it, but they gave it Apple and so on.

Fikes: Right.

Winograd: [01:22:21] But what they were doing then is visionary technologies that were going to be the future. "Now, let's do that again." Very explicitly not starting with "here's the answer to what we're going to find." "We have to figure out what it is." Liddle, of course, had been very active at Xerox PARC. He was the head of the development group that took the PARC computers and made them into products. He knew that environment totally; an extremely bright guy. He had co-taught with me in my HCI course at Stanford. I remember the main problem with that—it was great teaching with him except that because of his other job, he wanted to have the course at 8:00 in the morning, and the students didn't. [laughter] So we had to compromise on when to have the course, but I had worked with him there.

So Allen picked him as the person to bring together Interval. And he was basically going around, collecting people he thought would do interesting stuff. It wasn't highly targeted. It wasn't like, "Okay, we need experts in this." Having worked with him both from the early days of PARC but also when we taught together, he said, "You'd be really a good person to come for the startup." It was from the beginning clear that he wasn't saying, "You're the person we want here permanently." I mean, not

that he would have refused, but it wasn't what he was looking for. But as somebody who knew people, who was thinking about issues, who could try to really bring together something.

[01:23:59] That was the idea. I went there for, basically, the startup year and helped to bring in other people and set projects going. I had a workshop there on the web when the web was just starting. I can still remember, actually, the first time I saw a real browser was at Xerox PARC, where we brought had Mark Andreessen. He had been developing the Mosaic browser, and he brought it up, and there was this picture of dinosaurs and links to the Dinosaur Museum in Hawaii.

I was used to you type, you fetch a file, and then load the file. You could use the net. I was using the net or the internet, but the immediacy—and this is, again, human-computer interaction—of clicking on something and the picture shows up as opposed to downloading a picture file and showing it. I said, "This is going to be big." Now, I can't get credit for being the only one who said that. So part of the development there was on Internet stuff. So that's what I did for that year.

[01:25:19] Google was a very different thing. Larry came and gave a talk in my HCI seminar. We had a speaker series; we had several hundred speakers over the years. And it was about what he had learned about HCI doing Google. And at the end of the talk, I said, you really have a lot of great experience and learning from what you're doing, and I have sort of a theoretical background for it. We should write a book together.

He said, "Oh, that's interesting. Yeah, we should do that." So [he] said, again, step by step, "Why don't you come to Google for a year? We'll pay for you there. You can do it as a sabbatical, but we'll pay the full amount. And we can work together." Then, as it got closer, he said, "You know what? I'm pretty busy. I don't have time to work on a book, but why don't you come anyway?" So I got to Google with no real expectations about what I was going to work on because the book isn't happening.

I worked on a few projects. I was part of what was at the time called the Caribou Project, which became Gmail. It had already been going on. I didn't start it, but I was one of the people who got together in meetings to discuss what should be done. I worked on some ideas of my own for something which sort of looks like a—I don't know—it was a home brew version of stuff which later came out by other companies, not by Google.

[01:26:59] Then one of the big things was I talked to Jonathan Rosenberg, who was the product manager. And he said, "I have this problem. We like to hire these people called associate product manager, APMs, who are right out of school, mostly master's degrees, some of them PhDs, mostly master's, very smart. And they are interested, and their job is to integrate between the technical and the human part of it. So they work with the business development, they work with the interaction side, and with technologists. And their job is to pull those things together in a coherent way for a given project." The APM project at Google is famous. I mean, it was really copied by a lot of companies and so on. He said, "They

need guidance, and I can't give them all guidance. So could you be somebody to help guide them?"

I said, "Oh, you mean like advising grad students?"

He said, "Yeah, it's like advising grad students."

So I was the grad advisor, if you want to call it that, for several of the APMs. So I had dabbled in various other projects. And several of [the people I advised] have gone on to become major Silicon Valley people in their own right.

[01:28:29] I won't say I dabbled, but during my year there, I didn't have a focus. Sometimes somebody would say, "Here, can you look at this?" And, "Here's a project I'd be interested in your advice on." I went to meetings where they were discussing where to go in various directions.

Fikes: It does sound like your description of you working with your students, your PhD students, and the role that you played. Moving forward, you became emeritus in 2014. What was that decision about? What brought that up?

Winograd: You asked me that ahead of time, and I was trying to think. One of the things I said at the very beginning is my life has not been one of big, agonizing decisions, but rather, "Oh yeah, that's the right next step." If you say, how did I decide to go to MIT? Well, it's just the obvious thing to do. Or how did I decide to work in language? Well, it was the obvious thing to do. And at that point, I think various things happened. Josh Cohen, whom I taught this Africa course with, left Stanford. He went to Apple University, which is another whole interesting thing. So that course, I either had to find somebody different—and we had really worked on it very closely

together—or end that course. We had hired somebody else in HCI, so it was no longer I was *the* HCI professor. We had Scott Klemmer who is a very bright young guy and ready to take over things.

My need to be there, in some sense, was less because I did feel responsible for the program. I didn't want it to just drop dead if I left. I was not particularly happy being tied down to a teaching schedule. I think lots of people feel that at various times in their lives. If I wanted to go on a trip, or I wanted to do something or even just not be there on a Thursday, the teaching mechanism and the whole way of doing that was [tying me down].

[01:30:44] There were three courses I enjoyed teaching. One is the d.school course, which we've talked about. One was the course on the book I did with Flores called Phenomenological Foundations of Computing, Cognition, and Language. And it was a great course because it was really a philosophy course, not a computing course. I mean, it was about those things, but people read things and wrote papers and had discussions. And the other was an undergraduate first-year course, that, at one point, the university said, "We did a survey of what the undergraduates like and don't like about Stanford. One of the things that they don't like is they can go through four years at Stanford and never be in a classroom with a real professor and fewer than 100 other students. They got big lectures by the famous people, and then they had sections with the graduate students, but they never got a chance to just get to know a real professor." So they put incentives into place to get faculty to teach what they called freshman seminars and sophomore [college]. So you could get a full-course teaching

credit for teaching sixteen students sitting around a table in a freshman seminar and complete freedom as to the topic. They said to the faculty, do what you want. And there were people who brought in a guest speaker every week and did nothing else. [laughter] And it was a free lunch. I didn't do it that way. I had a few guest speakers, but I was more actively engaged.

But it was a great course. I loved it because, first of all, you were getting [enthusiastic] students. I taught in the fall, so they had just arrived at Stanford. So their attitude about learning and about what they needed to do for courses and so on was not shaped yet. So you could get them to think, not just do the exercises. And they were eager and bright. The topic was, basically, social issues of computing.

So I was able to bring that whole social issue side in. It was not a technical computing course. It was looking at the same kind of things we've talked about in the social issues. I still have roughly monthly meetings with a former student. I had him in that class the first quarter he arrived at Stanford, and I'm still basically advising him. He's now head of a company.

Fikes: Wonderful.

Winograd: [01:33:36] I had a couple of courses that I really liked, but that was not enough to fill the teaching requirement. So I would have had to make up a course. It just didn't seem like, okay, I'm ready to not have the course schedule. I don't feel like making up another course**.** I had brought in people to teach the HCI courses, so I could do another HCI course, but they were well under control. And Scott needed them to fill his teaching requirement.

Fikes: Right.

Winograd: So I didn't want to take them away from him. So at that point, I said, "Well, it's a good time to retire."

Fikes: Has that worked out well for you?

Winograd: Yeah, it's worked out great. So the other thing, which is incidental in some sense, but that was about twelve years ago, something like that. And my oldest grandchild is now fifteen. There have been four more since then, and they all live in the area. So if you say, what do you do with your time? I'll give you the local example. Tuesday, after I met with you, we went off and picked up our 12-year-old—he's the second one, second brother—from middle school, brought him here to the house, did stuff with him all afternoon, which included his playing all the stray band instruments we have stashed away, and took him out for dinner. And that was our afternoon.

[01:35:08] We try to meet with one of our grandsons—they're all boys—every week. That dropped dead during COVID because of the obvious, but we're trying to pick it up again because it was something we really liked. The twelve-year-old is still happy to spend time with us. The fifteen-year-old has gotten really interested in politics. So he and Carol can have endless discussions about the political scene and the candidates and all that's going on. So fortunately, even the older ones still enjoy being here. And the younger ones really do. They'll say to their mom, "When can we go see Bubbe and Grandpa?" So the stage of life I was getting into was

national boards, philanthropy, grandchildren, just sort of a natural what happens to you when you're over seventy.

Fikes: You really are spending more time with your family. [laughter] I mean, that's something that they often say when people are retiring. I wanted to ask you what accomplishments in your career have been most satisfying to you and the greatest impact?

Winograd: So I think there's, first of all, the obvious landmarks. SHRDLU was a major—what's the right word—kick in the pants, sort of spark plug. It got people excited about what was going on.

Fikes: Absolutely. I remember that.

Winograd: Even if I think that the direction we were going wasn't going to work, I think that having had that influence and getting people to think about it was great. The book with Flores was the next big one, and it was, how did you really rethink what was going to work and not going to work. It was more than just AI. It was how does computing fit into people's lives? I think that's had a lot of subtle influence. It's not that it's a theory you'd win a prize for, but it was something that people would step back and say, "Oh, yeah. I need to think about that."

[01:37:20] I think that's one of my satisfactions with teaching; that's when I think I've succeeded with a student. Not when they say, "I've got it!," but when they say, "Oh, yeah. I need to think about that." So in terms of visible things—publications and research and stuff—I think those two are the big marks. Fomenting things, so it's the things like CPSR and the ethics course. I mean, I did this ethics course at Stanford thirty years ago,

something like that, forty maybe, whatever it was. Then it got turned over to Eric. I see you talked to Eric Roberts. Then he left, and now they picked it up again. They said, "Okay, this is something we want, really, to teach." So I think, although I'm not involved in the latest iteration of that, I was part of the infrastructure, the background that led to it. I think bringing that into the computer curriculum, bringing that into what people think about—again it's what people think about.

[01:38:20] Along those lines, then there's the students. And my greatest satisfaction in teaching has always been connections to the students, not the research papers. If you look at my CV, for somebody of my age and status, I have a very small number of research papers. They're there, but it's not like I was churning them out at several a year, as a lot of my colleagues do because that was never what really motivated me. It was never what felt great. It was working with students and getting to know them and seeing how I could help their development and what they were thinking about.

The fact that some of them went on to do very big things is great, but even the ones who didn't do big things, I think it's great to have that influence on them and to be able to reflect back. I'll have people out of the blue say, "Oh, do you remember me? I was a student of yours back in so and so, and you really affected me. Your thinking was really important to me." Oh, great. That's nice. I didn't even think of that person. But they're there.

Fikes: That's wonderful. And you seem to have taken a very independent path at Stanford, sort of above the typical boundaries and directions that many of us followed. And that's thrilling to hear about that. And it strikes me that it relates to what you talked about growing up, that you were on your own, you were on your own path, you were not part of the group as much.

Winograd: When I step back and say, who am I? What makes me tick? I think that analysis is there.

Fikes: That's been exciting for that to come out in our talk here. Do you have any general reflections about Stanford? We've already talked about multiple department and interdepartmental stuff, but anything you want to say about Stanford and its history and where it's going?

Winograd: I mean, of course, Stanford history is a big topic, as you know. For me, it was absolutely the right place. I think we earlier mentioned interdisciplinarity. I mean, I think if I had stayed at MIT, which I could have, I would have been much more channeled into the next thing in the AI lab, as opposed to connections to design and all these other things.

[01:41:10] In the period I've been here, it's wealthy enough that you could do a lot of stuff without worrying about money. I never went in for big grants. I mean, this one I mentioned on digital libraries was the biggest, probably, and I was one of four faculty. And it was not even that big, by grant standards. I didn't build an empire. I never hired a research associate.

Fikes: Wow.

Winograd: I had grad students that I paid their grad thing, but that was it. And that was fine. I could do that and still function in the department. I think there are a

lot of places where, if you don't bring in the money to do things and hire research associates, you're not part of the game. I think there's two pieces of the students, the graduates and undergraduates. The undergraduates are very stimulating. They're smart. They're really bright kids and pretty open to things.

[01:42:19] MIT's a great school, but if you look at the MIT undergraduates, it's a different population. They're smart, too, but they're much more focused, whereas Stanford, you really get a variety, and you get different points of view. I'd say points of view is worth a lot. Then the grad students, because of Stanford's position—and it's grown more and more so over the years—it gets the best of the crop. I mean, you could argue CMU, Stanford, Berkeley. But, I mean, for all intents and purposes, we're getting the best computer science students there are, and that makes a huge difference.

I sometimes said if I were to be stuck teaching at Podunk State, could I even teach because I've always worked with these really bright students and eager and interested and so on. I don't know how to drag people along. If they need to be dragged along, I just let go. That's why they didn't pick me as an advisor. I don't know if I have that skill to do that. I just have such high expectations. And Stanford supports that. Stanford is a place where you can have super high expectations for the students, and that happens. It works.

[01:43:42] And the balance, compared to other tech schools. I think, for example, Hennessy did a very good job of saying, "Okay, I'm going to

be your first engineering president, but I'm not going to throw away the rest of the university. Humanities can come along, and they can all continue to develop and be supported well." I think that attitude is very good compared to, "Well, why are you bothering if it's not technology?" I have really very little complaint of any kind for the university. I mean, I can complain about individual people. We all can. But I think it's been a really positive place.

Fikes: Were you involved in directing the university in any sense, at the dean's level or with Hennessy when he was president, to effect what you just said, to bend that arc?

Winograd: Bend that arc, no. And so I think it's an interesting observation. And it comes from this outsider—I mean, the things we've talked about in terms of attitude—plus the attitudes people had toward HCI and towards my conversion from AI and all those things.

[01:44:59] So they didn't ask, but, I mean, to have gone through where I've gone through and never even become a department chair, you would say, well, he's sort of skipping out. I mean, everybody puts in their time to be a department chair, even if they're no good at running departments. I can name some.

Fikes: Yep.

Winograd: But it's just what you do. I think they certainly felt that if I were, that would be bad for our fundraising because that's not what I like to do. And if I say you don't want to take military funding, that's certainly not a popular thing to do if you're chair of the department.

Fikes: Right.

Winograd: So I was never on the Faculty Senate, never a department chair. I was head of the HCI group, but that was two people. [laughter] I just have not gotten involved in governance, except the d.school. I mean, I was sort of founding faculty of the d.school. But there, it was really more supportive of a central vision rather than, "Let's all get together and figure it out."

Fikes: Well, it's absolutely fascinating to me because, I mean, basically, you were out changing the world while these other things were going on.

Winograd: I mean, I have to admit, one of the things that made me ready to retire is I could stop going to faculty meetings.

Fikes: [laughs]

[End of Part 2 of 2 – Interview with Terry Winograd – February 16, 2023]

Terry Winograd

Curriculum Vitae

Education

1970 M.I.T — Ph.D. (Applied Mathematics)

1967 University College, London — (Linguistics)

1966 The Colorado College — B.A. (Mathematics)

Employment

1973-present, Computer Science Department, Stanford University

1973-74, Visiting Asst. Prof;

1974-79, Asst. Prof;

1979-89, Assoc. Prof.;

1989-2014 , Professor

2014-present, Professor Emeritus

2005-present, Founding faculty, Stanford d.school (Hasso Plattner Institute of Design at Stanford)

2002-2003: on leave from Stanford at Google

1992-93: on leave from Stanford, at Interval Research, Palo Alto.

1970-73 M.I.T. (Instructor in Mathematics and Asst. Prof. of Electrical Engineering)

Consulting (long-term)

Google, 2001- 2010

Interval Research, 1993-1998

Action Technologies, Inc., Alameda, CA., 1987- 1996

Xerox Palo Alto Research Center, 1972-1983

Awards

2011 ACM SIGCHI Lifetime Research Award

2008 Benezet Award, The Colorado College

2004 Elected to CHI Academy (ACM SIGCHI)

1999 Rigo Award for lifetime contributions to Computer Documentation (ACM SIGDOC)

1996 Founders Award, Computer Professionals for Social Responsibility

1987 *Understanding Computers and Cognition* named as the Best Information Science Book of 1987 by American Society for Information Science.

1986 Honorary D.Sc., The Colorado College.

1977 Mellon Junior Faculty Fellowship.

1977 Dean's award for excellence in teaching, Stanford.

Editorial and Advisory Boards

Advisory Boards:

Credo Reference

Google

IFixMe

K9 Ventures

Mobile Metrix

ReframeIt

Virtual Tribunal Project (Berkeley War Crimes Study Center)

Voices from the Rwanda Tribunal

Vsee Labs

USACM Policy Committee

Editorial Boards:

Informatica, Journal of Human-Computer Interaction, ACM Transactions on Human-Computer Interaction, User Modeling and User-Adapted Interaction

National Offices:

President, Computer Professionals for Social Responsibility, 1987-1990

National Board, Computer Professionals for Social Responsibility, 1984-1996

Memberships

Computer Professionals for Social Responsibility , Association for Computing Machinery, IEEE, ACM Special Interest Group on Computers and Society, ACM SIGCHI.

Publications

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Kumar, M., Garfinkel, T., Boneh, D., and Winograd, T. Reducing Shoulder-surfing by Using Gaze-based Password-entry, SOUPS: Symposium on Usable Privacy and Security. Pittsburgh, PA. 2007

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Interviewer Biography
Richard Fikes

Richard Fikes is a Professor Emeritus in Stanford's Computer Science Department and the former director of Stanford's Knowledge Systems Laboratory. Before coming to Stanford, he did seminal work in the development of automatic planning and plan execution monitoring systems for intelligent robots at SRI International's Artificial Intelligence Center, developed knowledge-based system technology at Xerox Palo Alto Research Center (PARC), and led research groups at IntelliCorp Inc. and Price Waterhouse Technology Centre.

At Stanford, Fikes was an innovative leader in the development of techniques for effectively representing and using knowledge in computer systems. He led projects focused on developing large-scale distributed repositories of reusable computer-interpretable knowledge, collaborative development of multi-use ontologies, enabling technology for the Semantic Web, reasoning methods applicable to large-scale knowledge bases, and knowledge-based technology for intelligence analysts. He was principal investigator of major projects for multiple Federal Government agencies including the Defense Advanced Research Projects Agency (DARPA) and the Intelligence Community's Advanced Research and Development Activity (ARDA).

Since retiring from Stanford, Fikes has been professionally active as a consultant on design and development of knowledge representation and reasoning technology, an expert witness in cases involving disputed patents of Artificial Intelligence technology, and a mentor of high school students learning how to do research.